Deloitte.

Report on Assam Solar Roadmap

January 2024

Asian Development Bank

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List of Abbreviations

AEGCL: Assam Electricity Grid Corporation Limited
AERC: Assam Electricity Regulatory Commission
ALMM: Approved List of Models and Manufacturers
APDCL: Assam Power Distribution Company Limited
APGCL: Assam Power Generation Corporation Limited
BESS: Battery Energy Storage System
BOO: Build, Own, Operate
BOOT: Build, Own, Operate, Transfer
CAGR: Compound Annual Growth Rate
CEA: Central Electricity Authority
COD: Commercial Operation Date
CPSU: Central Public Sector Undertaking
DISCOM: Distribution Company
DRE: Distributed Renewable Energy
DSM: Demand Side Management
EPC: Engineering, Procurement, and Construction
ESO: Energy Storage Obligation
ESS: Energy Storage System
FSP: Floating Solar Project
FY: Financial Year (In the current context it is 1-April till 31-March)
GDAM: Green Day-Ahead Market
GTAM: Green Term-Ahead Market
GW: Gigawatt
GWh: Gigawatt hours
HPO: Hydro Purchase Obligation
IPP: Independent Power Producer
ISTS: Interstate Transmission System
KSEB: Kerala State Electricity Board
KW: Kilowatt
MMBT: Million Metric British Thermal Units
MOU: Memorandum of Understanding
MVA: Mega Volt-Ampere
MW: Megawatt
OEM: Original Equipment Manufacturer
PPA: Power Purchase Agreement
PPP: Public-Private Partnership
PSP: Pumped Storage Plant
REC: Renewable Energy Certificate
RES: Renewable Energy Source
RESCO: Renewable Energy Service Company
RPO: Renewable Purchase Obligation

SLDC: State Load Dispatch Center
SPPD: Solar Power Park Developer
TBCB: Tariff-Based Competitive Bidding
USD: United States Dollar
WPO: Wind Power Purchase Obligation

Executive Summary

Background

The North-Eastern Region in India with special emphasis to Assam has emerged as a hub of economic activity over the last decade. Assam's Gross Domestic Product (GDP) has grown from INR 1,569 billion to INR 4,932 billion between FY[1]13 and FY23 at a Compounded Annual Growth Rate (CAGR) of ~12%. As a natural consequence, demand for electricity has also grown from 6,495 GWh in FY13 to 11,481 GWh in FY23 (~6% CAGR in the past 10 years; higher than national growth rate and north-east region) fueled by other factors like increase in number of consumers, heatwaves, and lifestyle related changes in the domestic consumers.

To cater to the current demand, Assam has been relying on a thermal dominated procurement portfolio wherein ~68% of the energy demand is met from coal and gas-based generation sources. The State's installed capacity is 1,993 MW in FY23 with peak demand of 2,376 MW leading the state to procure power from high-cost power markets to meet its shortfall in supply. Of the 1,993 MW installed capacity in the State, only 21% of the capacity belongs to the state-owned Generation Company APGCL with remaining 68% and 11% owned by Central Sector Generating Companies and Independent Power Producers. The State Electricity Distribution Company APDCL is also facing the issue of rising power procurement costs where in the average power procurement costs (APPC) has risen from INR 4.19 per kWh in FY22 to INR 5.38 per kWh backed by rising fuel costs for gas and coal based generating stations and power market prices. The APPC for Assam is higher than states like Tripura, Meghalaya, Manipur, West Bengal, Jharkhand, Bihar etc. Although this highlights APDCL's endeavor to meet the required energy demand at any cost which is laudable in its own right; however, with increasing trend of projected demand (energy requirement of 21,619 GWh and peak demand of 4478 MW by FY32) and increased requirement of energy supply capacity, the current power procurement strategy may not be sustainable in the long run.

Push for Renewable Energy

To push for greater use of clean energy in the consumption, the Central Government had setup Renewable Purchase Obligations (RPO) for electricity distribution companies (or Discoms) in India which requires the Discoms to meet a percentage of their energy supply requirements through renewable sources. It is implemented in each state by the State Electricity Regulatory Commissions, which in Assam is the Assam Electricity Regulatory Commission (AERC). As per the targets setup by the Central Government in July 2022[2] and revised targets setup in October 2023 till FY 2030, the state electricity distribution company, Assam Power Distribution Company Limited (APDCL) would require to source ~43% of its energy requirement through Wind, Hydro, Distributed Energy Resource (DER) Source and other Renewable Energy Sources by FY2030[3]. This target may expand to 47% by FY2032[4].

Solar Capacity Addition Requirement

In terms of Renewable Energy (RE) potential, as per Central Government estimates, Assam has a potential of around 14.24 GW of possible RE implementation in state. Out of this, solar potential makes up for 13.76 GW followed by small hydro and bio-power at 202 MW and 279 MW respectively. The State has no potential for Wind Energy and has been resorting to buy wind power from outside state. Thus, to meet the RPO targets, specifically for the Distributed Energy Resource (DER) and Other Category, Assam will have to rely heavily on solar power. By FY32, the state would require 4.2 GW of installed solar capacity.

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[1] FY refers to Financial Year (1-April till 31-March)
[2] Source: Ministry of Power, Government of India
[3] Source: Source: Ministry of Power, Government of India
[4] Note: While the Ministry of Power, Govt. of India has provided RPO targets till FY30, an escalation assumption of increase in RPO target between FY29 and FY30 has been assumed to continue till FY32 for the purpose of this analysis

The Government of Assam's Renewable Energy (RE) Policy for 2022 aims to add 1200 MW (1000 MW from solar and 200 MW from other sources) of renewable energy by 2027. Additionally, the government plans to install another 1000 MW of solar power on unused state government land. To align with this policy, address Assam's growing electricity demand and meet the RPO targets, the Assam Power Generation Company Limited (APGCL) and the Assam Power Distribution Company Limited (APDCL) aim to increase capacity by approximately 4.2 GW by the fiscal year 2030 (highly likely capacity of 3.5 GW[5]). Considering the planned capacity addition till FY32, in the initial years (FY23 to FY25), there is a consistent shortfall in planned solar installed capacity compared to the required capacity. The shortfalls range from 251 MW to 486 MW, indicating the need for accelerated efforts to meet the solar capacity targets. Between FY26 to FY28, there is a surplus in planned solar installed capacity compared to the required capacity. This surplus ranges from 176 MW to 453 MW, showcasing successful efforts in exceeding the solar capacity targets during this period. FY29 and FY30 experience small shortfalls, while FY31 and FY32 show substantial shortfalls, emphasizing the need for strategic adjustments to align with long-term targets. There is a need for significant capacity addition of 1.5 GW by FY32 additional solar capacity (over the planned capacity) to bridge the widening shortfalls.

Prevalent Business Models to Implement Utility Scale Solar Project

To meet the long-term targets, the assessment of prevalent business models for Grid-Connected Utility Scale Solar Projects reveals several viable options, each with its own set of advantages and challenges.

Public Sector Owned Utility Scale Solar Model: This model involves the government utility handling the entire lifecycle of the solar project, from development to operation. It is suitable for small to large-scale projects and is funded through equity from the state government/utility, along with project-specific debt from commercial banks/financial institutions. The regulator determines the tariff on a cost-plus basis and a competitive selection of EPC contractors. This model offers simplicity in transaction structuring and allows the utility to build expertise for future projects. However, challenges may arise in the utility's ability to develop and efficiently manage solar projects, given limited experience.

CPSU Scheme for Solar Projects: The Central Public Sector Undertaking (CPSU) scheme allows central and state PSUs to execute solar projects with Viability Gap Funding (VGF). The government entities participate in tenders, and the tariff is capped with the bid's basis being the quantum of VGF quoted. The government/utility provides equity funding, and project-specific debt is raised from commercial banks/financial institutions. Viability Gap Funding is released in two tranches, and the tariff is capped to promote affordability. This scheme simplifies transaction structuring and contributes to the development of utility expertise. However, due to capping of VGF and pre-determined tariff, utilities in north-eastern India may find this scheme unattractive due to unique terrain and development challenges.

Private Owned Utility Scale Solar Models (BOO and BOOT). Build Own Operate (BOO) Model: In the BOO model, the project developer takes complete responsibility for the project, including land and connectivity. The ownership rests with the private developer, and the project is suitable for large grid-connected solar projects. The financing involves equity from private developers and project-specific debt from commercial banks/financial institutions. The BOO model allows for cost optimization through tariff-based competitive bidding, attracting private players with niche expertise. However, delays may occur if dependence on other agencies for clearances is present.

Build Own Operate Transfer (BOOT): Solar Park Model: In BOOT model, solar projects are established in pre-identified lands with clear demarcations, and the responsibility of land and connectivity lies with the Solar Park Implementing Agency (SPIA). The developer signs a PPA with an intermediary procurer, and after 25 years, assets are transferred to the SPIA. The BOOT model offers

[5] This includes 2.4 GW of Solar, 0.65 GW of Thermal and 0.4 GW of Hydro based generation capacity.

ready land and basic infrastructure at the outset, resulting in a shorter turnaround time for project execution. However, the implementation can be complex due to numerous transactions involved.

Each business model presents unique advantages and challenges. The Public Sector Owned Utility Scale Solar Model could be beneficial for building local expertise, but the government's limited experience in managing solar projects may pose challenges. The CPSU Scheme simplifies transaction structuring but requires careful consideration to avoid inefficiencies. Private-owned models (BOO and BOOT) offer cost optimization and attract private players, but they may face delays and complexities in certain situations. Assam needs to carefully evaluate these models based on its specific requirements, resource availability, and administrative capabilities to choose the most suitable approach for sustainable solar project implementation.

Way Forward

Assam has faced challenges in implementing Independent Power Producer (IPP) and Public-Private Partnership (PPP) projects. Issues such as difficult terrain, high transportation costs, and power evacuation challenges have deterred private players, resulting in the realization of only a modest 70 MW capacity at a single location.

In response to these challenges and the huge requirement of 4.2 GW of solar capacity by FY2032, it becomes imperative to expedite the establishment of renewable energy projects in Assam, particularly at a large scale, to meet the surging demand and fulfill RPO requirements. The state utilities' support is crucial in initiating successful demonstrable large-scale projects. While private participation can be incorporated, a utility-led capital expenditure (capex) model with initial public utility responsibilities is envisioned. This strategic approach aims to address existing implementation challenges, such as land and evacuation issues, and lay the foundation for a gradual transition to PPP and eventually IPP models in the state.

The phased approach, beginning with a utility-led capex model, ensures that the state utilities play a pivotal role in the early stages, taking on significant responsibilities. This involvement not only helps overcome initial hurdles but also establishes a framework for future large-scale projects and the achievement of market maturity in the state. As major capacity projects under the utility-led model prove successful, private participation can be gradually increased, leading to a natural evolution towards more private sector-led models. This transition allows for the gradual transfer of risks from the public to private players, fostering a sustainable and progressive development trajectory for renewable energy projects in Assam. Overall, this strategic and phased approach aims to create a conducive environment for the successful execution of large-scale projects, paving the way for Assam to meet its renewable energy targets and contribute substantially to the country's clean energy goals.

2 Introduction

2.1 Background

Over the past decade, the North-Eastern Region of India, with a special emphasis on Assam, has witnessed remarkable economic development. Assam's Net State Domestic Product (NSDP) has grown at a Compound Annual Growth Rate (CAGR) of approximately 11.2% from Financial Year (FY) 12 to FY 22, surpassing the national average of 10.7%[6]. Simultaneously, the demand for electricity in Assam has steadily increased at a CAGR of around 5.9% during the same period, exceeding national average of 3.9%. Assam's strategic location near neighboring countries like Bangladesh, Bhutan, Nepal, China, and Myanmar presents unique opportunities for economic, strategic, and cultural collaborations in the vast Asia-Pacific region.

Despite these successes, Assam faces energy security challenges. Historically, the State has relied on gas and hydro-based power generation, facing challenges related to gas availability and price fluctuations. The Government of Assam is committed to promoting renewable energy, introducing initiatives like the Assam Renewable Energy (RE) Policy in 2022 with a target of adding 1,200 Mega-Watt (MW) of renewable energy capacity. Additionally, schemes like "Mukhya Mantri Souro Shakti Prokolpo" aim to install 1000 MW of solar projects on government-owned land, signaling a strong shift towards a greener energy portfolio.

The Ministry of New and Renewable Energy (MNRE), Government of India, estimates the solar power potential of approximately 13.7 Gigawatt (GW) in Assam, far higher than the current installed capacity of around 264 MW. To harness this significant potential effectively and ensure synergy among all stakeholders, a phased solar roadmap for the state is imperative. This roadmap will foster coherence and create an enabling environment for large-scale solar projects in Assam, contributing to economically viable and sustainable energy solutions.

2.2 Approach to development of solar roadmap

The solar roadmap has been developed considering the overall historical trends, future outlook of the sector and c the energy transition plans which the country and state has embarked on. They key policy enablers at the national and state level have been analyzed and key areas of interventions identified based on the best practices. A critical input required for the solar roadmap is the demand trajectory in the state which needs to be managed. The CEA projections for Assam prepared as part of 20th Electric Power Survey (EPS) report have been taken as the base and appropriate corrections in the assumptions applied based on updated demand supply scenario. The demand assessment also includes the requirements on account of forecasted growth in the electric vehicles (EV) and additions due to implementation of green hydrogen policy. On the supply side, the existing and future generation additions have been identified in consultation with state utilities.

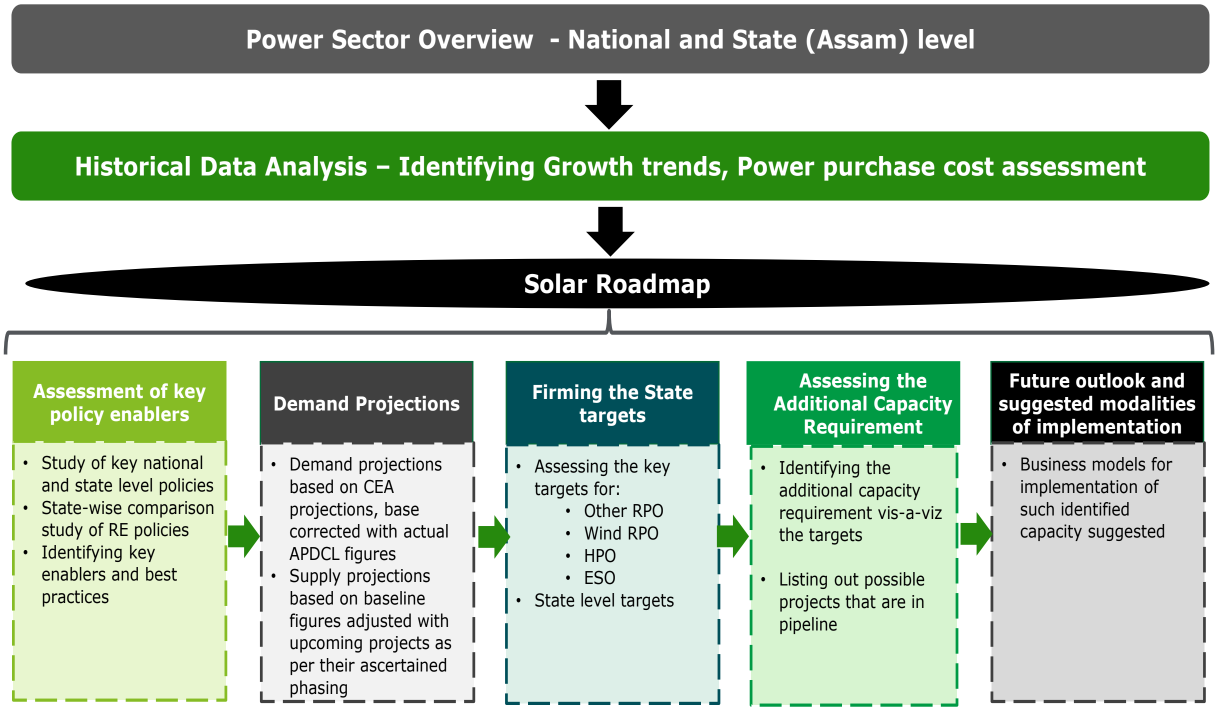
The renewable purchase obligation (RPO) targets at the national and state level have been assessed to prepare the technology-wise breakup of renewable portfolio for the state. In addition, the resource wise renewable potential available in the state has been analyzed to evaluate the optimal renewable mix available in the state. The annual generation portfolio has been prepared and the solar capacity requirements worked out until 2030.

A consultation workshop was organized in September 2023, where key stakeholders from across the value chain attended the workshop and shared their insights. The approach to the study and results were presented to all the stakeholders. The workshop explored effective business models tailored to Assam's needs and scenarios, inviting the private sector to provide perspectives on successful project development models and address challenges in developing utility-scale solar projects. Best practices in financing and procurement of solar power projects were also

[6] Source: Ministry of Statistics and Program Implementation, Government of India (MOSPI)

presented, incorporating valuable feedback from stakeholders. The list of attendees is attached in the Annexure 9.11

Figure 1: Our Approach towards the Solar Roadmap



Consultation Workshop to corroborate the study approach and results

The approach of the study and the preliminary results were presented to all the key stakeholders in Assam in a consultant workshop. The feedback received from the stakeholders have been incorporated in the report. The list of attendees of the workshop is attached in Annexure 9.11

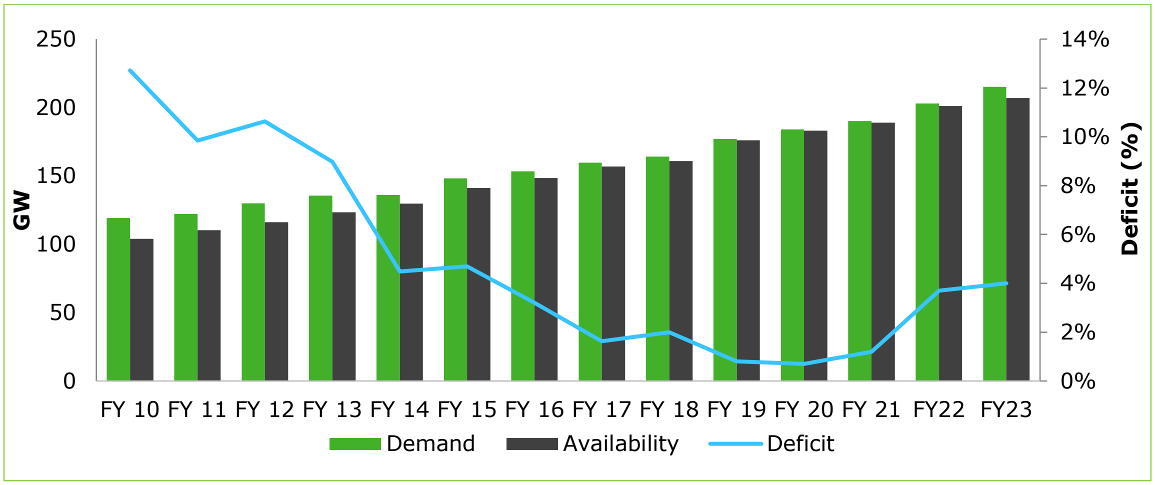
3 Power Sector Scenario in India

3.1 Key trends in the Indian power sector

The electricity demand in India has risen at compound annual growth rate (CAGR) of 4.7 percent in the last 10 Years (FY13 to FY23) backed by rapid electrification, growth in per capita electricity consumption and Government's push for increased industrialization through schemes such as "Make in India" and "Production-Linked Incentive" (PLI) incentive programs.

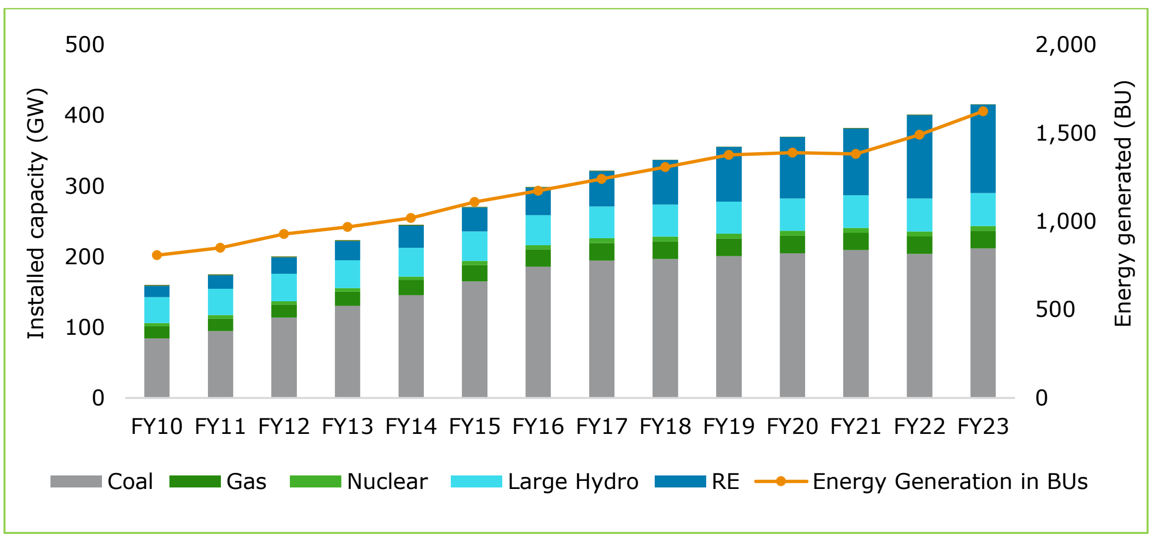
3.1.1 National Demand Supply Scenario:

Figure 2: Demand Supply status in India



To meet this rising demand, the supply capacity has also increased at a CAGR of around 5.1 percent which has helped to narrow the energy deficit (earlier ranged between 9–12 percent) to around 4 percent in FY23.

Figure 3: Source wise Installed Capacity (GW) and energy generation (BU)



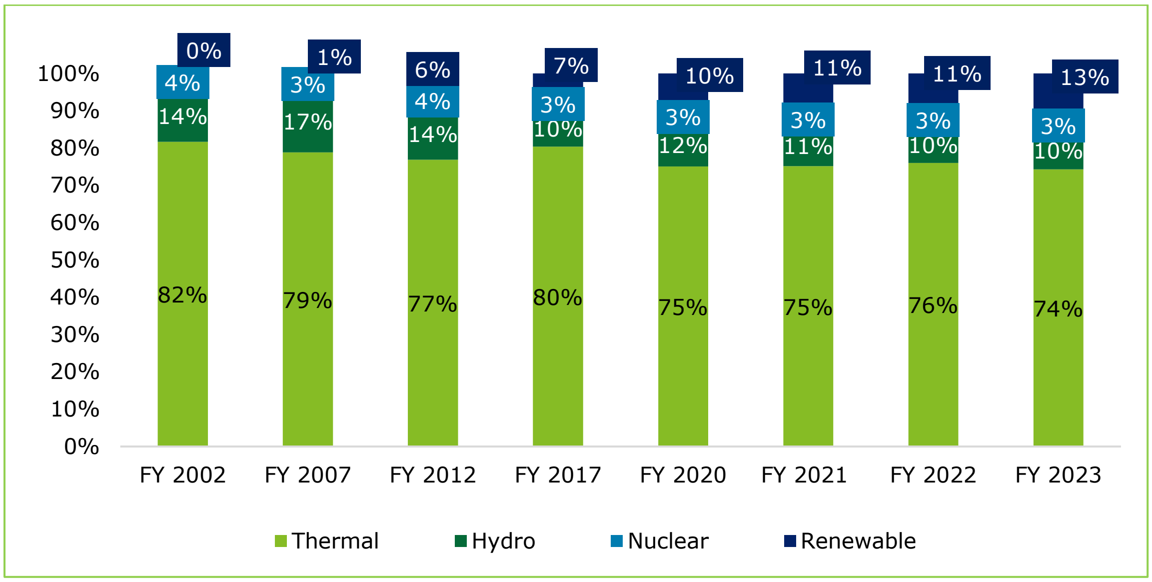
Historically India had been dependent on thermal based power generation which continues to support the base load requirement of the country. About 60% of the total installed capacity comprises coal, gas and nuclear based generation in FY23. Meanwhile, share of RE in installed capacity in overall energy basket of India has increased steadily from 10% in 2010 to reach 30% in 2023, driven by the Government of India's ambition to add 175 GW of RE by 2022 which has now been expanded to 450 GW by FY30. RE addition has happened at CAGR of ~17% (FY10-23) outperforming all other energy sources during the same period.

Both the Public Sector (Central / State) and Private Sector have been instrumental in meeting the rapidly growing electricity demand in the country. The capacity additions which were dominated by public sector in the early 2010's has witnessed a remarkable shift with private sector outperforming capacity additions by Central and State sector entities. Private sector now contributes ~50% of total installed capacity in FY 23[7].

Share of Generation across Sources (%):

The increase in RE capacity has led to a steady increase in the share of RE generation which increased from 6 percent in 2012 to 13 percent in FY 2023. Meanwhile, the share of thermal has come down marginally from 77% to 74% during the same period.

Figure 4: Source wise share of generation



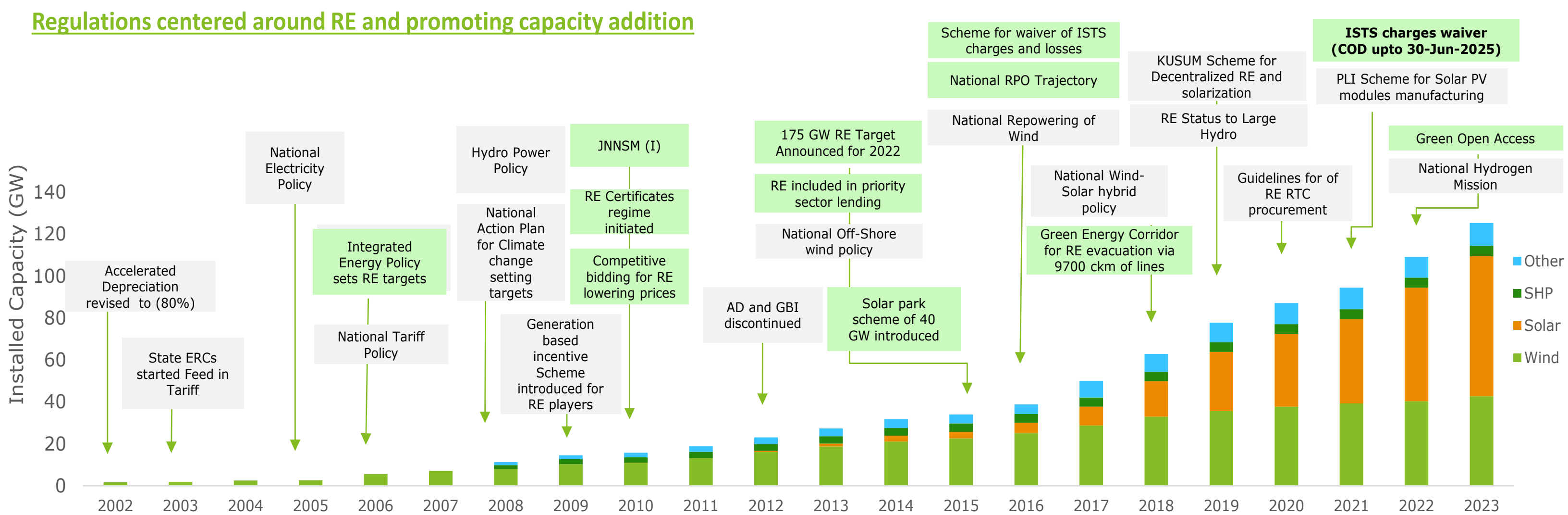
[7] Source: CEA

3.2 Renewable Energy Sector in India

The rapid greening of the power sector in India has been led by the policy & regulatory reforms which has pushed for accelerated capacity development for Solar and Wind based power projects over the past two decades. This has been further driven by global decarbonization and energy transition push as well as national commitments in the form of Nationally Determined Contributions (NDC) and Net Zero goals of India.

The key regulations centered around renewable energy and promoting clean energy capacity addition in the country have been enumerated below[8]:

Figure 5: Key developments in renewable sector in India



Global factors & National Commitments	Policy Initiatives	Regulatory Initiatives
• GoI has scaled up targets to install 450 GW of Renewable power by 2030 • India's INDC commitment – Reduce Emission intensity of its GDP by 33-35% from 2005 • Stringent emission norms for thermal plants.	• **29 States/UTs** have their own State Renewable Policies • **23 States/UTs** have dedicated Solar Policies • **8 States/UTs** have dedicated Wind Policies	• Ministry of Power in 2021 have set national RPO targets till FY 30; all states are required to align their own state RPO targets with same • Waiver on ISTS charges for Solar, Wind, ESS, Hydro PSP ESS having COD before 30th June 2025 (31st Dec 2030 for Green Hydrogen and Green Ammonia projects)

[8] MOP, MNRE, CEA

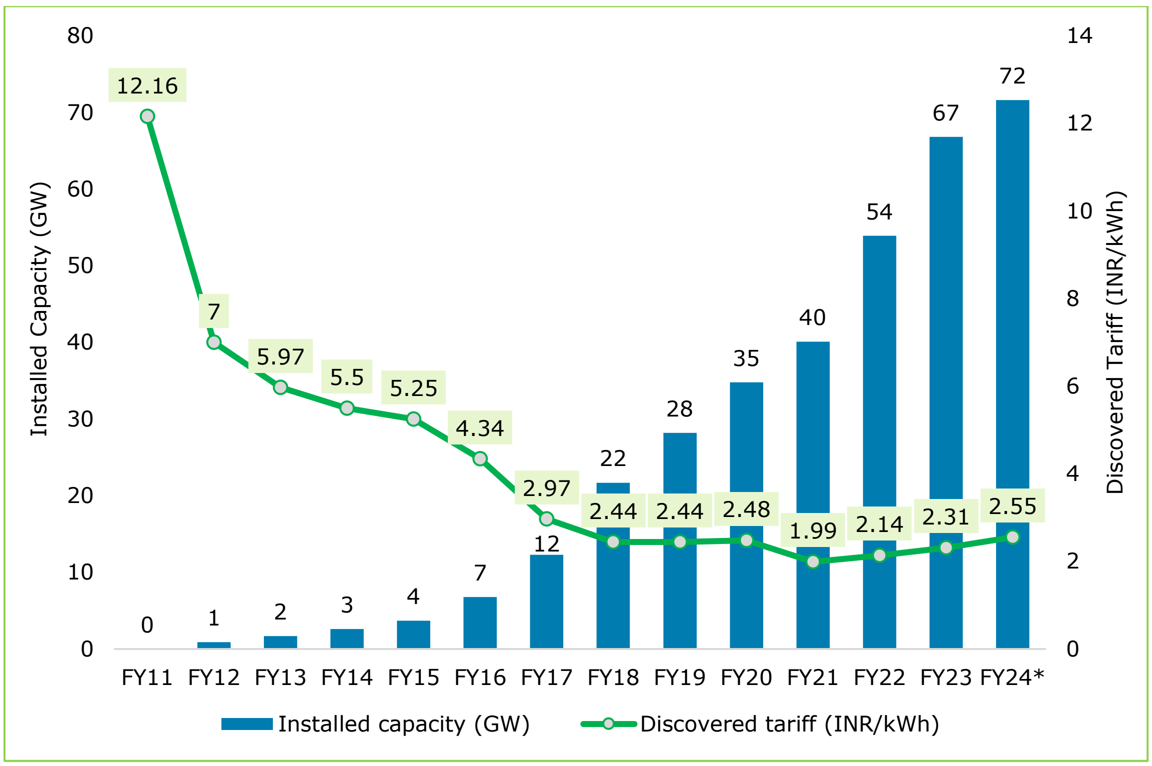
3.3 Solar sector in India

Solar installations have witnessed a CAGR of ~33% between FY 17 and FY 23 which have led to fall in solar tariffs in the country due to large economies of scale and is further fueling the growth as tariffs have fallen below the cost of thermal based generation. Reduction in solar tariffs during earlier years was majorly due to mandated renewable power purchase obligations driving competition in supply, reduction in module prices, and favorable rates of financing. Now with many local firms setting up domestic manufacturing units for modules, increasing competition between Solar OEMs and aggregation of solar bids by Solar Energy Corporation of India (SECI), the solar tariffs had even fallen to at a record low of INR1.99 per kWh in competitively bid out projects in FY21.

While there has been a slight increase in tariffs in the last 2 years due to higher module costs and customs duty on equipment, the country continues to add 13-14 GW of solar capacity annually.

3.3.1 Solar capacity growth and lowest discovered tariff trends[9]:

Figure 6: Solar Capacity Addition vs Lowest Tariff Discovered



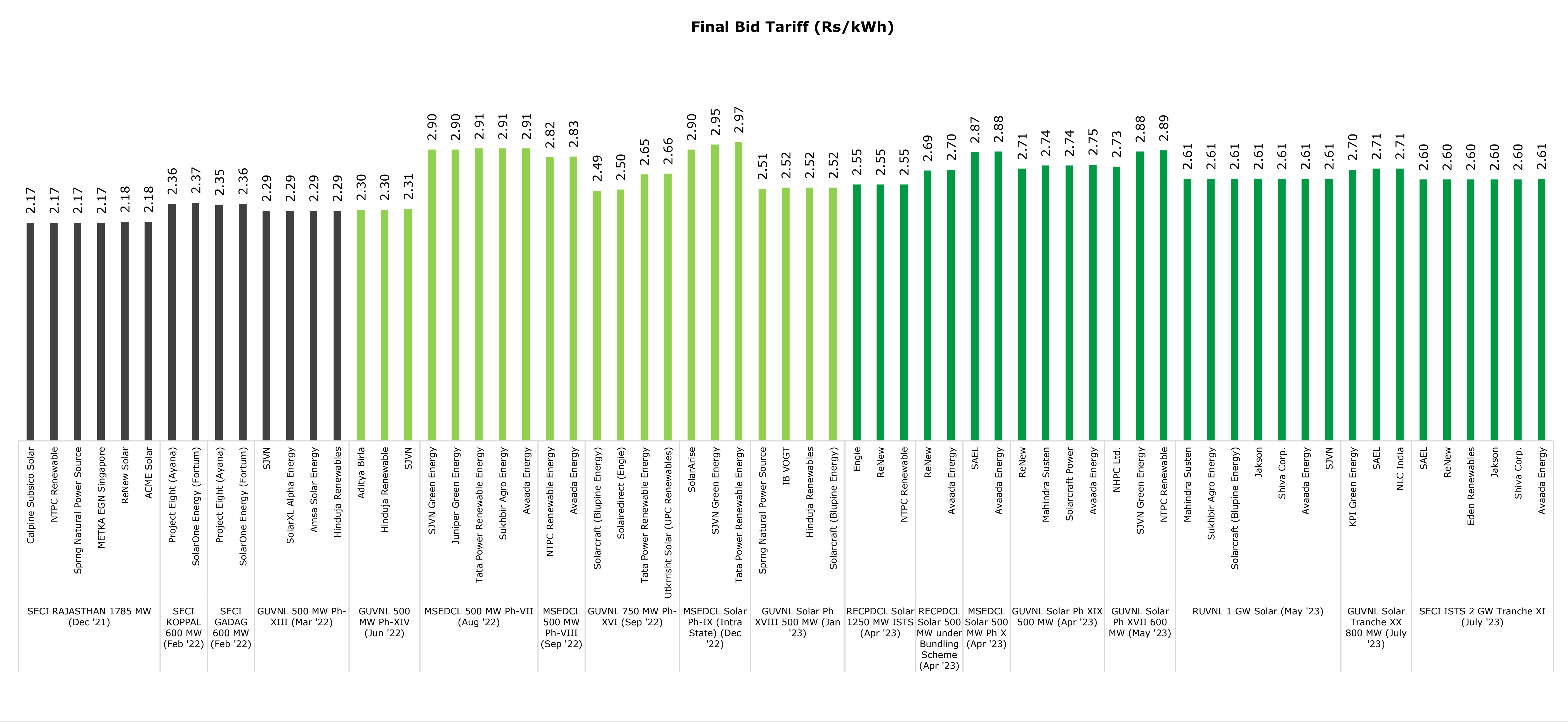
*Till August 2023

[9] CEA and Deloitte Research

Solar tariff trends:

Some of the historical successful solar project bids in competitively bid-out projects have been illustrated below:

Figure 7:Solar Tariff Trend (Tariff Based Competitive Bidding)



4 Assam Power Sector: Status and Performance

4.1 Power Sector Framework

The State's power sector went through reforms in 2004, wherein the erstwhile Assam State Electricity Board (ASEB) was restructured vide Government of Assam Memo No PEL.151/2003/Pt./165 dated 10th Dec' 2004 into following five entities to carry out functions of generation, transmission and distribution of electricity:

a) Assam Power Generation Corporation Limited (APGCL) (Generating Company)

b) Assam Electricity Grid Corporation Limited (AEGCL) (Transmission Company)

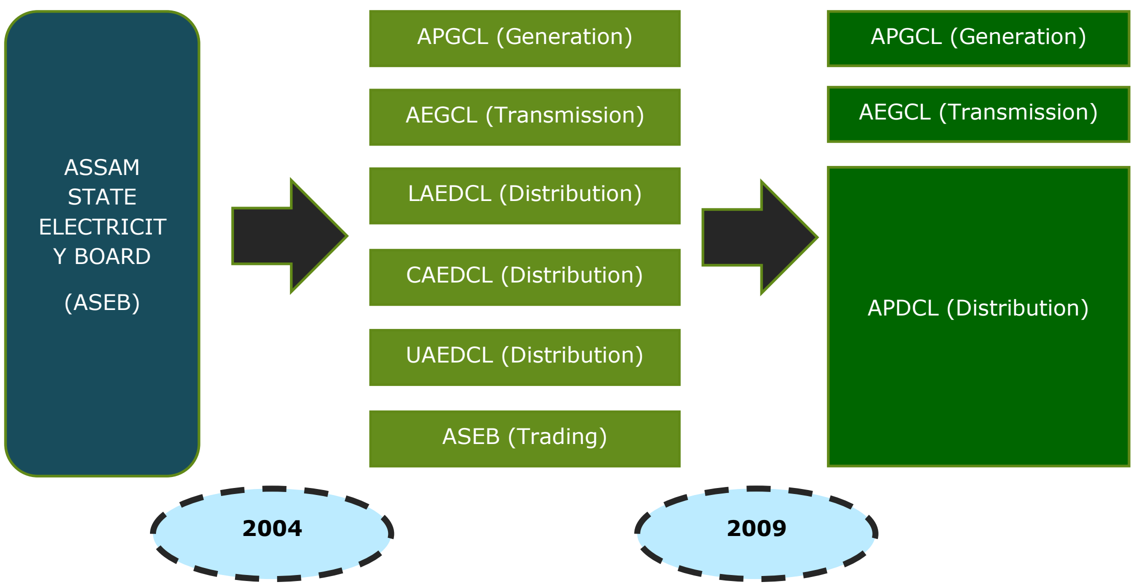
c) Lower Assam Electricity Distribution Company Limited (LAEDCL) (Distribution Company)

d) Central Assam Electricity Distribution Company Limited (CAEDCL) (Distribution Company)

e) Upper Assam Electricity Distribution Company Limited (UAEDCL) (Distribution Company)

In May 2009, as per GOA notification No PEL.41/2006/199 dated 13th May 2009, as per Assam State Reform (Transfer and merger of Distribution Functions and undertakings) Scheme, 2009 CAEDCL & UAEDCL Distribution Company merged with the LAEDCL thereby forming a single distribution company for the State. Further, in October 2009, the name of the company was changed from LAEDCL to Assam Power Distribution Company Limited (APDCL) vide Certificate of Incorporation dated 23rd October'2009[10].

Figure 8: Reforms in the Assam Power Sector



The institutional setup, roles, responsibilities and financial metrics of the present public sector entities in the Assam Power Sector have been enumerated in Annexure - 9.1.Institutional Setup and Financial Performance of the Public Entities in Assam Power Sector

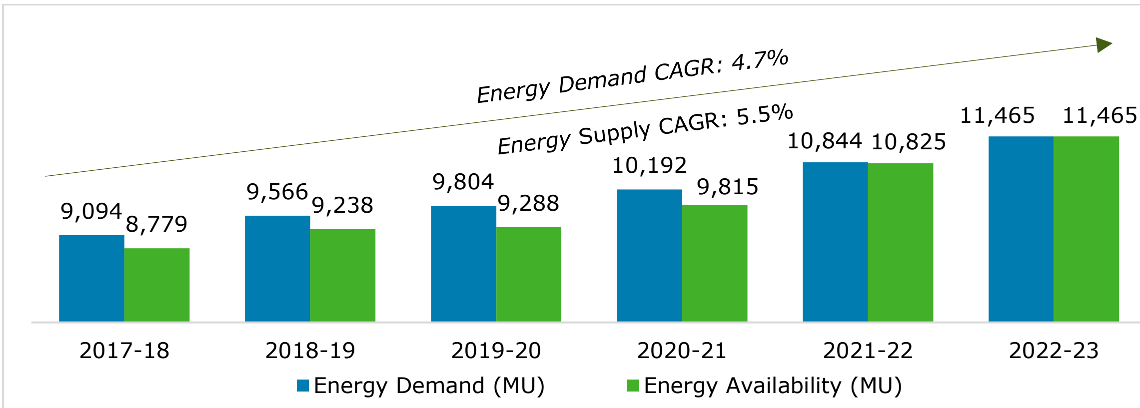
[10] Joint initiative report on Power for All from Govt. of India and Govt. of Assam (https://powermin.gov.in/sites/default/files/uploads/joint_initiative_of_govt_of_india_and_assam.pdf)

4.2 Demand Supply Scenario in Assam:

Power Demand in Assam has witnessed a rapid growth of CAGR of ~4.7 percent between FY18 to FY23 increasing from 9,094 GWh to ~11,465 GWh respectively, outperforming the entire Northeastern Region demand growth of ~3.3% CAGR during the same period. The State's energy supply which was in deficit of ~3.4% in FY18 has now achieved parity with demand in FY23 through mainly short-term power purchases which have come at higher costs.

4.2.1 Energy Demand and Availability in Assam:

Figure 9: Energy Requirement vs Supply in Assam

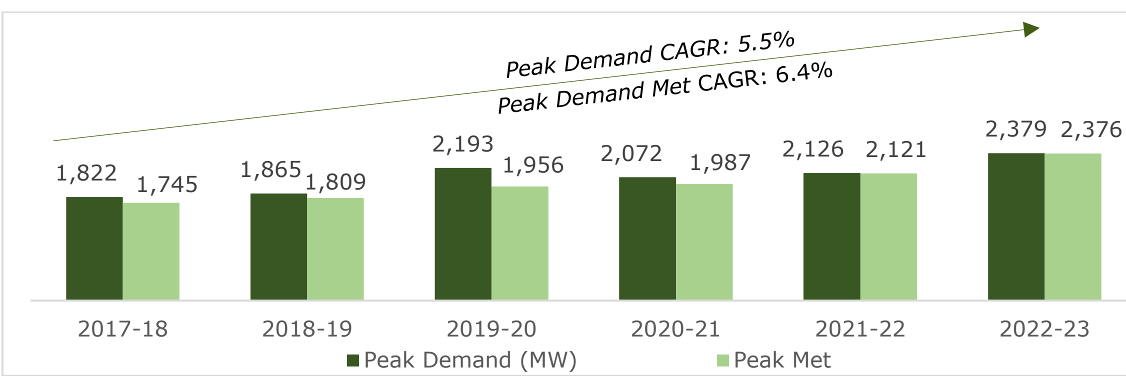


In FY 24, Energy requirement has already reached ~5,731 GWh from April to August 2023 (first 5 months of the financial year)[11].

4.2.2 Peak Demand and Peak Demand Met in Assam (MW):

The peak energy demand of the State has grown at a higher rate of 5.5 percent CAGR rising from 1,822 MW in FY18 to 2,379 MW in FY23[12]. Similarly, to meet this rapid growth, supply capacity has also grown at a CAGR of ~6.4 percent to meet the historical deficits and reach parity with demand.

Figure 10: Peak Demand vs Met (MW; Assam)



In FY 24, Peak Demand as on August 2023 has already reached 2,535 MW.

[11] CEA Monthly Power Supply Position Reports (Energy)
[12] CEA Monthly Power Supply Position Reports (Peak)

4.2.3 Supply Scenario in Assam:

The Installed Capacity in Assam has grown by 168 MW during the period FY 2019 – FY 2023. Coal and Natural Gas dominate the installed capacity of the State with a total share of 68% followed by Hydro (18%) and Solar (12%). The solar capacity has grown over five times from 41 MW to 248 MW during the same period. With the peak demand reaching ~2,500 MW in FY24 (till August) the State has been meeting its demand from short term power sources as well as Central Allocations.

4.2.3.1 Installed Capacity available to Assam (MW)[13]

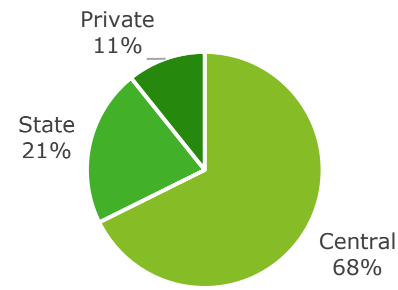
Table 1: Installed Capacity of Assam (MW)

Resource-wise Capacity	FY 19	FY 20	FY 21	FY 22	FY 23	% Contribution (FY23)
Coal	750	750	750	750	750	38%
Natural Gas	644	644	645	620	600	30%
Hydro	350	350	350	350	350	18%
Solar	41	43	69	148	248	12%
Other Renewable Energy Sources (RES)	40	42	42	45	45	2%
Total	**1,825**	**1,829**	**1,856**	**1,913**	**1,993**	**100%**

4.2.3.2 Installed Capacity Share in FY 23:

In terms of capacity ownership, Central allocations contribute ~68% of the State's available capacity (in MW terms) followed by State's own share of 21% and remaining by private developers at 11%. Assam Power Generation Corporation Ltd (State Genco) has a total electricity generating capacity of ~422 MW and is dominated by gas plants (309 MW). The private share is mostly in Solar Capacity with a share of around ~203 MW of capacity.

Figure 11: Installed Capacity share w.r.t Ownership

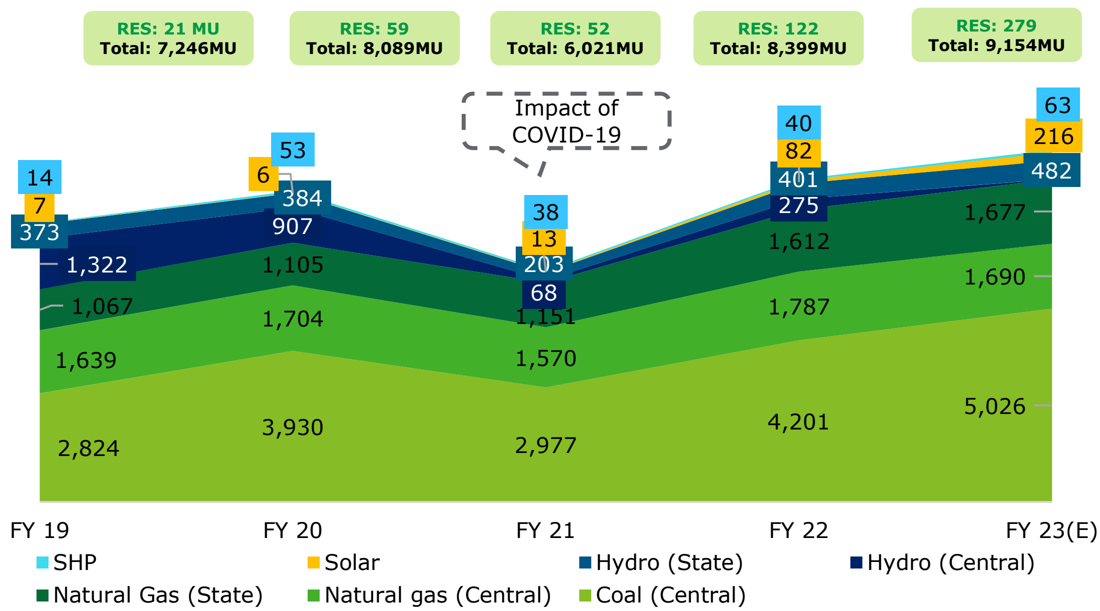


4.2.3.3 Energy Generation in Assam:

The energy generation in Assam has grown from 7,246 GWh to 9,154 GWh during the period FY 2019-2023. The fossil fuels such as Coal and natural gas contributes around 55% and 37% respectively to the overall generation mix. The renewable energy sources (RES) and hydro contributes to only 8% in energy terms.

[13] Source: CEA and APDCL data

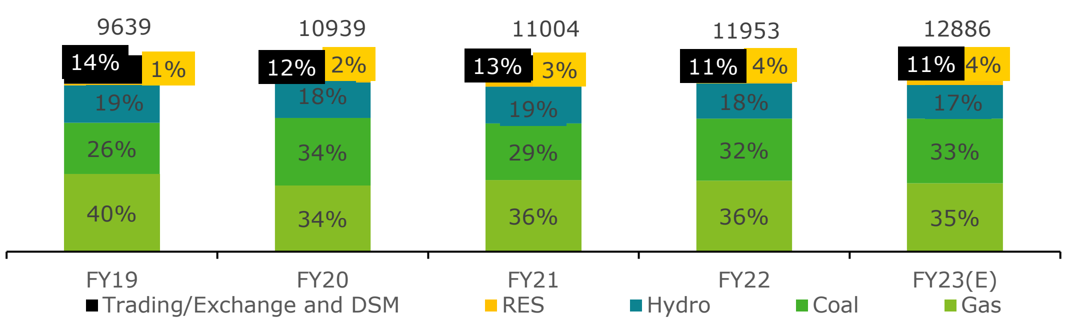
Figure 12: Energy Generation in Assam (GWh)



4.2.3.4 APDCL's Power Purchase Mix:

To cater to state's electricity demand, Assam has allocations in the central sector generating stations (CSGS) and also relies on the direct procurement through other sources like traders, bilateral agreements and power exchanges.

Figure 13: Energy Purchase Mix of APDCL (MU)



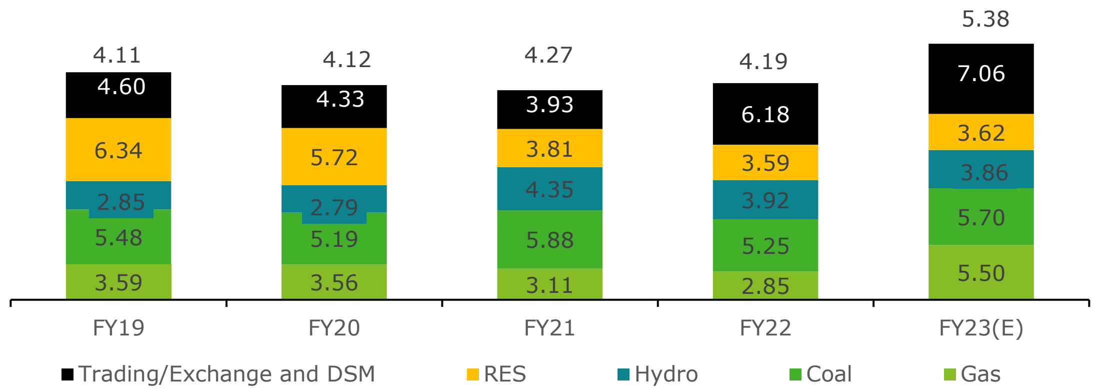
The share of coal in the power purchase portfolio has risen from 26% in FY19 to 33% in FY23(E) due to constraints in capacity expansion of gas plants (availability of gas been challenging) and limited increase in Renewable Energy Sources (RES) based capacities during the same period.

4.2.3.5 APDCL's Power Procurement Costs

The power purchase costs from gas and trading activities has increased significantly in FY23 which has led to the power purchase costs rising to nearly Rs 5.4 / KWh from historical levels of FY19 to FY22[14].

[14] Source: APR Order FY23 - APDCL

Figure 14: Average Power Purchase Cost (Rs / kWh)

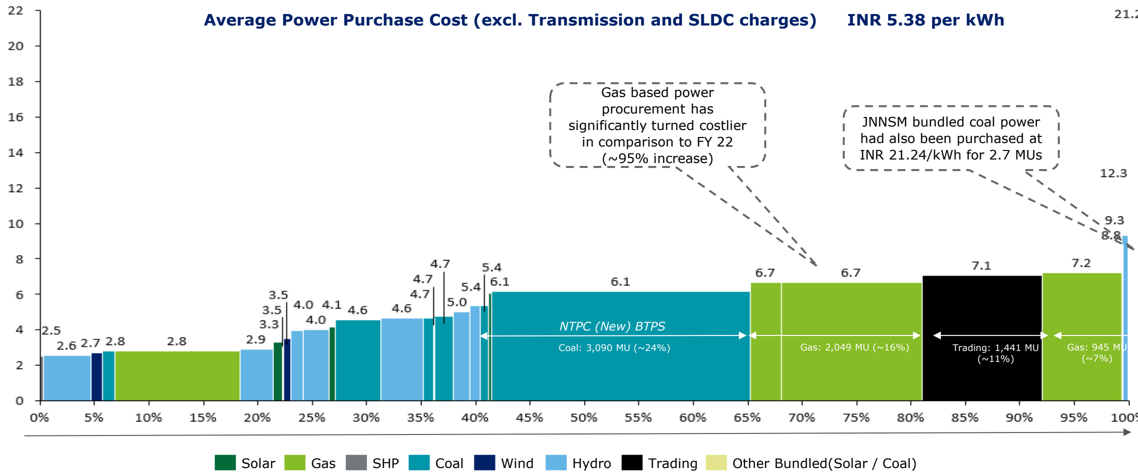


The average power procurement from gas-based plants has increased from INR 2.85 per kWh in FY22 to INR 5.50 in FY23 due to the significant increase in natural gas prices in India. This was majorly due to the geo-political situation affecting global fuel prices which were historically linked to the domestic gas pricing formula adopted by the Govt. of India. While North-East Region gets a price discount of 40% on the natural gas pricing (under Administered Price Mechanism), the prices increased by over 200% between FY22 and FY23. From FY 24 onwards, Govt. has approved floor price of US $4/ Million British Thermal Units (MMBTu) and ceiling price of US $6.5/MMBTu for next two years.

Figure 15: Half-yearly gas price comparison - FY 22 and FY 23 (USD / MMBTu)

Similarly, the power purchase cost from trading / DSM (Deviation Settlement Mechanism) segment also witnessed a significant rise from average cost of INR 3.93 per kWh in FY21 to INR 7.06 per kWh in FY23 which has been due to the surge in national power demand driving short term prices coupled with increasing fuel prices. Even coal-based power has been costing an average tariff of over INR 5.50 per kWh which is driving the average power purchase cost for APDCL.

Figure 16: Merit Order Dispatch - FY 23 (Estimated)



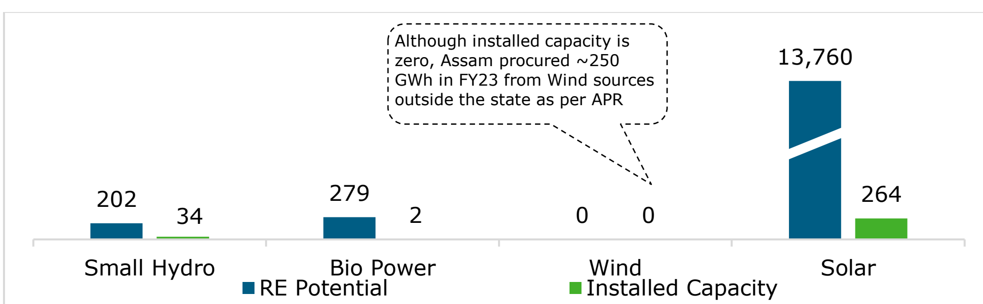
As can be seen from the graph above (based on total power purchase cost for APDCL APR Order for FY23), about ~60% of the total power purchase quantum comprising of coal, gas and trading-based power is being bought at over INR 6 per kWh creating financial challenges for APDCL which in turn impacts the retail tariffs for the State.

4.3 Renewable Energy in Assam

As per MNRE, Assam has a potential of around 14.24 GW of possible RE implementation in state. Out of this, solar potential makes up for 13.76 GW. However, with respect to actual implementation, only 248 MW solar capacity addition has been achieved till date. The other RE implemented has been hydro including SHP (42.6 MW). Major capacity addition of solar power in state has been through the Private Sector. Multiple implementation models have been executed in Assam, which include IPP, BOO, and Captive uses.

4.3.1 RE Potential vs Installed RE Capacity (Aug 2023):

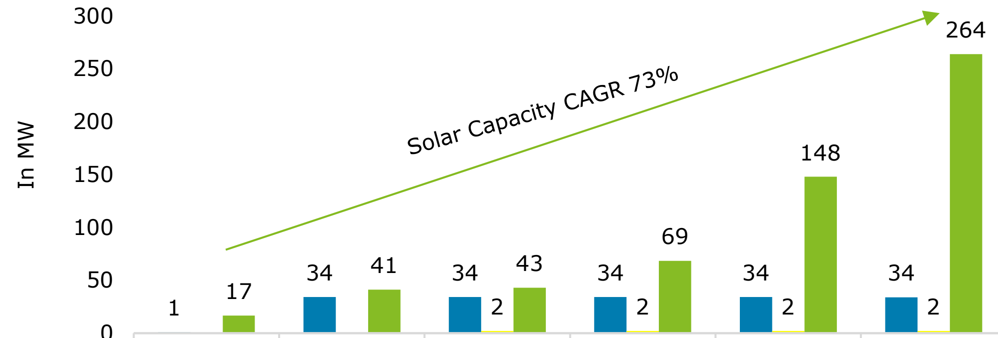
Figure 17: RE potential vs Installed Capacity (MW) in Assam



The potential for RE is 14 GW, out of which 95% is solar. There is limited scope to increase gas and coal based generation in the State. Issues ranging from environmental/ social perspectives, gas production expansion challenge and difficulty in setting up coal supply infrastructure continue to affect plans for capacity addition for thermal power generation. Thus, it is imperative to develop the available renewable potential in the State to meet the rising demand and ensure clean energy capacity proliferation.

4.3.2 RE Installed Capacity Trend in Assam:

Figure 18: YoY RE Installed Capacity Trend



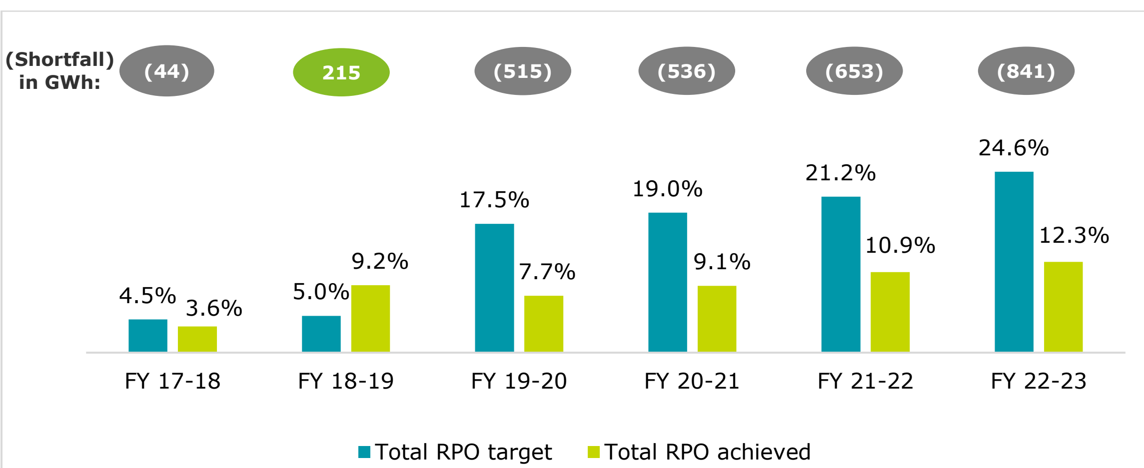
	2018	2019	2020	2021	2022	2023 (As on August)
Small Hydro	1	34	34	34	34	34
Bio-Power	0	0	2	2	2	2
Solar	17	41	43	69	148	264

In terms of RE installations, the State has witnessed increasing participation from the industry and APDCL / APGCL to setup Solar capacity which has resulted in ~5x increase in Solar Capacity since FY19. Small Hydro too has seen addition of 34 MW in the last 5 years.

4.3.3 Renewable Purchase Obligation (RPO):

Although the RE capacity has increased in recent years, the same is not sufficient to meet the Renewable Purchase Obligations (RPO) targets set by AERC[15]. The RPOs have largely remained unachieved due to low RE potential in the State and the high cost of building RE projects due to geographic and technical challenges.

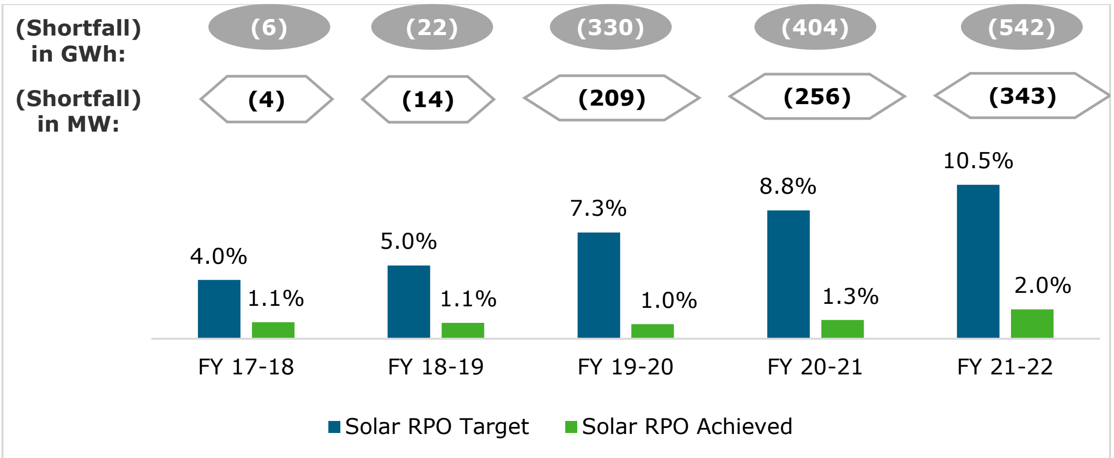
Figure 19: RPO Target vs Achievement – Assam



In terms of the Solar RPO targets, the Solar Capacity Gap had reached 343 MW in FY23.

[15] AERC, APDCL *Note: To calculate solar shortfall in MW, CUF of 18% has been considered

Figure 20: Solar RPO Target and Achievement[16]



From FY23 onwards, AERC has adopted the revised RPO targets set by the Ministry of Power which are steeper than the trajectory adopted by AERC till FY22, the impact of which has been discussed in subsequent sections. While, the Assam Electricity Regulatory Commission has not imposed any monetary penalty on APDCL for not fulfilling RPO obligations till now, it may impose penalty in future. The AERC (Renewable Purchase Obligation and its Compliance) Regulations, 2010 provides for effect of default by obligated entities such as APDCL on non-compliance with RPO. The relevant provision from the regulation is reproduced below:

"Clause 7.1

*If the obligated entities does not fulfill the renewable purchase obligation as provided in these regulations during any year and also does not purchase the certificates, **the Commission may direct the obligated entity to deposit into a separate fund,** to be created and maintained by such obligated entity, such amount as the Commission may determine on the basis of the shortfall in units of RPO and the forbearance price decided by the Central Commission:*

Provided that the fund so created shall be utilized as may be directed by the Commission for purchase of the certificates.

Provided further that the Commission may direct the State Agency to empower an officer to operate the fund and procure from the power exchange the required number of certificates to the extent of the shortfall in the fulfillment of the obligations, out of the amount in the fund.

*Provided also that **the obligated entities shall be in breach of its license condition if it fails to deposit the amount** directed by the Commission within 15 days of the communication of such direction.*

Clause 7.2

*Where any obligated entity fails to comply with the obligation to purchase the required percentage of power from renewable' energy sources or the renewable energy certificates**, it shall also be liable for penalty as may be decided by the Commission** under section 142 of the Act:*

*Provided that in case of genuine difficulty in complying with the renewable purchase obligation because of non-availability of certificates, **the obligated entity can approach the Commission for carry forward of compliance requirement to the next year**:*

Provided further that where the Commission has consented to the carry forward of compliance requirement, the provision of clause (7.1) of the Regulation or the provision of section 142 of the Act shall not be invoked."

Based on the above provisions, it is imperative for APDCL to meet the RPO obligations and tie up solar capacity to meet the requirements under RPO regulations for Assam.

[16] Note: For Determination of Solar Shortfall in MW terms, a CUF of 18% is considered over the shortfall in GWh.

5 Policy and Regulatory Framework

5.1 National level policies and guidelines

The Government of India has enacted several policy and regulatory measures which have led to the development of the power sector and give an impetus to renewable energy proliferation in the country. These have been the key drivers for triggering energy transition in the country which have also acted as a catalyst for shaping policies at the state level. Some of the national and state-level policies and regulations that have a significant impact on the renewable energy sector in India, particularly in Assam, are presented below:

Table 2: National Level Policies and Guidelines[17]

Sl No	Policy/ Regulation/ Schemes	Description
1.	National Electricity Policy, 2005	• Provides guidelines for accelerated development in the power sector and emphasizes energy security, positively influencing the renewable energy sector
2.	National Tariff Policy 2016	• Focuses on "4 Es (Electricity for all, Efficiency to ensure affordable tariffs, Environment for a sustainable future, Ease of doing business)" including Environment, leading to a sustainable future, which supports the growth of renewable energy
3.	National Electricity Plan, 2023	• Guides short-term and long-term demand forecasts and suggests areas for capacity additions, playing a crucial role in planning renewable energy projects
4.	Electricity Amendment rules, 2022	• Amendments, including specifications for timely recovery of power purchase costs and consideration of Energy Storage Systems, enhance the regulatory framework for renewable energy integration
5.	Waiver of Inter State Transmission System (ISTS) charges on transmission of electricity generated from Solar, Wind, Pumped Storage Plants, and Battery energy Storage Systems (2021)	• Provides financial incentives by waiving transmission charges, boosting investment and development in solar, wind, and storage projects
6.	Renewable Purchase Obligations (RPOs) and Energy Storage obligations Trajectory till 2029-30	• Specifies RPO trajectory, promoting the consumption of renewable energy and stimulating the growth of renewable projects. Year wise targets are set for different RE technology types include solar, wind, hydro etc.

[17] Deloitte Research

Sl No	Policy/ Regulation/ Schemes	Description
7.	Scheme for flexibility in Generation and Scheduling of Thermal/ Hydro Power Stations through bundling with Renewable Energy and Storage Power (2022)	• Encourages the integration of renewable energy with thermal power, offering flexibility and supporting Renewable Purchase Obligation (RPO) compliance.
8.	Guidelines for Tariff Based Competitive Bidding Process for Procurement of Power from Grid Connected RE Power Projects	• Facilitates transparent procurement of electricity from renewable energy projects, promoting competitiveness and efficiency in the sector
9.	Guidelines for Procurement and Utilization of Battery Energy Storage Systems	• Facilitates the integration of Battery Energy Storage Systems, enhancing grid support, flexibility, and stability

The national-level policies and guidelines introduced by the Government of India plays a pivotal role in fostering the accelerated development of the renewable energy sector, with a specific emphasis on solar power. The National Electricity Policy of 2005 provides crucial guidelines for the overall power sector, with a positive influence on renewable energy, emphasizing energy security. The National Tariff Policy of 2016 focuses on creating a sustainable future, providing a conducive environment for the growth of renewable energy through its "4 Es" approach. The National Electricity Plan, 2023 guides demand forecasts and capacity additions, crucial in shaping renewable energy projects. Amendments in the Electricity Rules 2022, including provisions for the timely recovery of power purchase costs and the consideration of Energy Storage Systems, enhance the regulatory framework for seamless integration of renewable energy. The waiver of ISTS charges on transmission and the trajectory for Renewable Purchase Obligations (RPOs) till 2029-30 provide financial incentives and promote the consumption of renewable energy, specifically boosting solar and wind projects. Additionally, schemes like the flexibility in generation and scheduling through bundling with renewable energy, and guidelines for transparent competitive bidding and utilization of Battery Energy Storage Systems, further underscore the government's commitment to fostering renewable energy, particularly in the solar power domain, ensuring grid stability, flexibility, and efficiency in the pursuit of sustainable and clean energy sources.

5.2 State level policies and guidelines

In line with national level policy push, Assam has also charted a vision through its RE policy 2022, to promote RE sources and set targets for capacity addition for the next five years. AERC, the state electricity regulatory commission, While AERC had already adopted the RPO targets set in July 2022, it is also expected to adopt the revised targets set in October 2023. Overall, major schemes, regulations and policies in Assam are provided below.

Table 3: State level policies and guidelines[18]

Sl No	Policy/ Regulation/ Schemes	Description
1.	Assam Renewable Energy Policy 2022	• Sets a target for additional renewable energy capacity, promoting solar park development and supporting the growth of renewable energy in the state. Key policy targets include: ○ Target of 1200 MW of additional RE power capacity by 2027 ○ Provision for setting up of 1000 MW solar plants on free unused government land as per "Mukhya Mantri Souro Shakti Prokolpo". ○ Solar Park development promoted; APDCL to act as Solar Park Project Developer ○ 250 MW assured offtake by APDCL for solar projects set up to supply solely to APDCL
2.	AERC (Terms and Conditions for Tariff Determination from Renewable Energy Sources), Regulations 2017	• Provides a regulatory framework for determining tariffs, ensuring fair compensation for renewable energy projects
3.	Assam Electricity Regulatory Commission (Renewable Purchase Obligation and its Compliance) Regulations '2010, (Third Amendment), 2021	• Expands the scope of RES, including Large Hydro and Pumped Storage, and introduces flexibility in meeting Renewable Purchase Obligation (RPO)
4.	AERC (Grid Interactive Solar PV Systems) Regulations, 2019	• Establishes capacity targets, standards, and safety requirements for grid interactive solar projects, contributing to the state's solar energy development
5.	AERC (Forecasting, Scheduling, Deviation Settlement and Related Matters of Solar and Wind Generation Sources) Regulations, 2018	• Facilitates grid integration of solar and wind energy while ensuring grid stability and security through forecasting and scheduling mechanisms
6.	AERC (Terms and Conditions for Open Access) Regulations 2018	• Promotes ease of doing business by facilitating Open Access, encouraging private investment in renewable energy projects
7.	Grid-connected Rooftop Solar Power Plant Programme (Feb 2018)	• Encourages the development of grid-connected rooftop solar projects, providing financial assistance and supporting the state's solar capacity
8.	Assam State Action Plan on Climate Change	• Investigates strategies to enhance resilience against climate change, aligning with sustainable and climate-friendly renewable energy practices

The state-level policies and regulations in Assam reflect a concerted effort by the government and regulatory authorities to prioritize and promote renewable energy, with a particular emphasis on solar power. The Assam Renewable Energy Policy 2022 establishes a target for additional renewable energy capacity, focusing on the development of solar parks and supporting overall growth in the state's renewable energy sector.

[18] AERC, Power Department

Figure 21: Salient Features of Assam RE Policy 2022

Target Capacity - Target of **1200 MW** RE capacity addition under RE policy period (i.e. by 31st Mar 2027) of which Grid Connected Solar Projects (620 MW), Grid Connected Solar Rooftop Projects (300 MW), Off-grid Solar Applications (80 MW) and other RE sources (200 MW)

Solar Power Park Developer (SPPD) - APDCL has been given responsibility to act as SPPD for development of infrastructure and managerment of Solar Park in State

Selection of Solar Power Generators (SPGs) and iscovery of corresponding tariffs etc. shall be done by APDCL and the latter would enter into long term PPA of 25 years with SPGs with the State Government to evolve a guaranteed payment security mechanism to the investors

Solar Plants for sale of power to Discoms other than APDCL - Deemed Open Access Provision as well as provision for execution of wheeling and banking agreement with Discom under captive mode have been provided for

Development of Solar Plants under REC mechanism - SLDC Assam designated as nodal agency for accredition of solar projects under REC mechanism

Other Provisions - Provisions related to **Captive Solar Plants, setting up of RE based EV Charging Stations, promotion of Pumped Power Storage Projects, Biomass, Waste-to-energy etc** have been provided for.

The detailed features of the Assam RE Policy 2022 have been enumerated in Annexure 9.2.

Regulatory frameworks, such as the AERC (Terms and Conditions for Tariff Determination from Renewable Energy Sources) Regulations 2017[19] and the AERC (Renewable Purchase Obligation and its Compliance) Regulations '2010 (Third Amendment) 2021, ensure fair compensation for renewable energy projects and expand the scope of renewable sources, including Large Hydro and Pumped Storage. The AERC's regulations on Grid Interactive Solar PV Systems (2019) and Forecasting, Scheduling, Deviation Settlement, and Related Matters of Solar and Wind Generation Sources (2018) contribute to the state's solar energy development by establishing capacity targets, safety standards, and mechanisms for grid integration and stability. Additionally, policies promoting Open Access, such as the AERC (Terms and Conditions for Open Access) Regulations 2018, encourage private investment in renewable energy projects. The Grid-connected Rooftop Solar Power Plant Program (Feb 2018) further supports solar capacity by incentivizing the development of rooftop solar projects. The Assam State Action Plan on Climate Change underscores the state's commitment to enhancing resilience against climate change through sustainable and climate-friendly practices, aligning with the promotion of renewable energy, particularly solar power, as a key component of Assam's energy strategy. These enabling policies collectively signify a strategic and comprehensive approach by the government and regulatory bodies in Assam towards advancing renewable energy, emphasizing the pivotal role of solar power in achieving sustainability and resilience goals. A detailed comparison of the RE policies and Ease of Doing Business in various Indian States has been enumerated in Annexure 9.4.

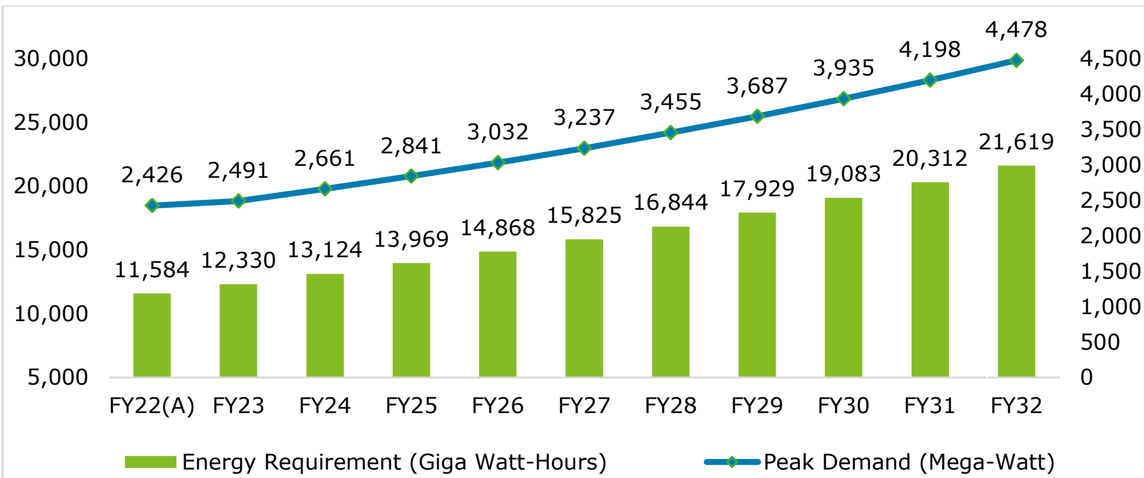
[19] Refer Annexure 9.3

6 Solar Power Roadmap / Capacity Requirement for Assam

6.1 Electricity Demand Forecast

The National Electricity Plan[20] formulated by the Central Planning Agency for Electricity (Central Electricity Authority-CEA) in India has projected that the electricity demand in Assam is expected to nearly double by the Financial Year (FY) 2032[21] from current levels. The electricity demand is being driven by the economic development of the region which is helping growth across major electricity consumer categories such as domestic, commercial and irrigation among others with a Compound Annual Growth Rate (CAGR) of 7-8% for each respective category. While the National Electricity Plan projected the demand from FY2022 onwards, actual electricity demand for FY2022 were incorporated to revise the projections till FY2032.

Figure 22: Base corrected demand projections in line with CEA's considered rate of growth[22]



The electricity demand is also being driven by the emergence of clean energy technologies and push in the form of electric vehicles and green hydrogen. As per the Electricity Plan, it is expected that about 49 million Battery-Electric Vehicles would be plying in India by 2032. Similarly, India has targeted to reach a production target of 10 million tons of green hydrogen by 2030. These targets if materialized may add about 2,000 Giga Watt-Hour (GWh)[23] of demand in addition to the electricity demand forecast for Assam. However, since these targets are dependent on factors such as technology maturity, incentives push from Government, investments from private sector, we have only considered the base electricity demand (21,619 GWh) forecasted by the Central Electricity Authority to materialize by FY2032.

[20] The National Electricity Plan along with forecast for demand in each State of India is prepared by the Central Electricity Authority and is called the Electric Power Survey (EPS). The last published EPS report is the 20th EPS published in November 2022. The EPS is aligned to the Long Term Generation Plan of the country which is also published by CEA.

[21] The Indian Financial Year follows an April to March period. For e.g., Financial Year 2032 (FY32) implies the period between 1st April 2031 to 31st March 2032.

[22] Source: CEA, Deloitte Analysis; Note: Energy Requirement has been base corrected basis actual achieved in FY22 and escalated based on Year on Year growth projection rates in 20th EPS for Assam.

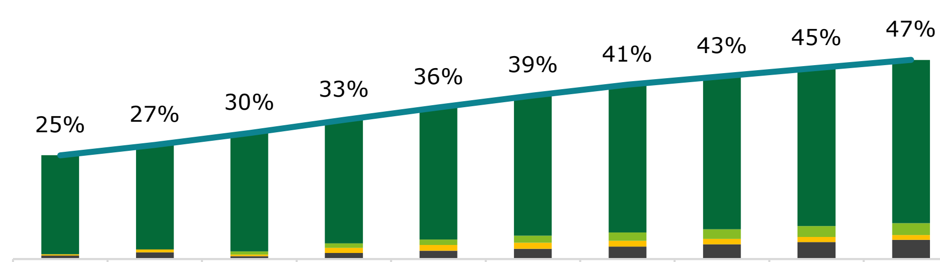
[23] Demand from Electric Vehicles (EV) and Green Hydrogen has been interpolated for Assam based on the demand for All India Electricity Demand from EV and Green Hydrogen adopted by CEA.

6.2 Renewable Energy Capacity Requirements

6.2.1 Renewable Purchase Obligations (RPO) Targets

To push for greater use of clean energy in the consumption, the Central Government had setup Renewable Purchase Obligations (RPO) for electricity distribution companies (or Discoms) in India which requires the Discoms to meet a percentage of their energy supply requirements through renewable sources. It is implemented in each state by the State Electricity Regulatory Commissions, which in Assam is the Assam Electricity Regulatory Commission (AERC). As per the targets setup by the Central Government in July 2022[24] and revised targets setup in October 2023 till FY 2030, the state electricity distribution company, Assam Power Distribution Company Limited (APDCL) would require to source ~43% of its energy requirement through Wind, Hydro, Distributed Energy Resource (DER) Source and other Renewable Energy Sources by FY2030[25]. This target may expand to 47% by FY2032[26]. While AERC had already adopted the targets set in July 2022, it is also expected to adopt the revised targets set in October 2023.

Figure 23: RPO Targets as per revised Ministry of Power guidelines (October 2023)



	FY 23	FY 24	FY 25	FY 26	FY 27	FY 28	FY 29	FY 30	FY 31	FY 32
Other RPO	23%	25%	28%	29%	31%	33%	35%	36%	37%	39%
DER RPO			1%	1%	1%	2%	2%	2%	3%	3%
Hydro RPO	0%	1%	0%	1%	1%	1%	1%	1%	1%	1%
Wind RPO	1%	2%	1%	1%	2%	2%	3%	3%	4%	5%
Total RPO	25%	27%	30%	33%	36%	39%	41%	43%	45%	47%

It is to be noted that historically APDCL hasn't been able to meet its RPO targets set by AERC. While no adverse action has been taken by AERC so far, the regulator may impose financial penalty as per the guiding regulations in the future.

6.2.2 Estimated Energy Requirement as per RPO Targets

The RPO targets for Assam would translate into the energy requirement as shown in Table below.

Table 4: Energy Requirements for RPO Target in Assam

S. No	Particulars	Unit	FY23	FY24	FY25	FY26	FY27	FY28	FY29	FY30	FY31	FY32
1	Assam Electricity Demand	GWh	12,330	13,124	13,969	14,868	15,825	16,844	17,929	19,083	20,312	21,619
RPO Requirements												

[24] Source: https://powermin.gov.in/sites/default/files/Renewable_Purchase_Obligation_and_Energy_Storage_Obligation_Trajectory_till_2029_30.pdf; https://cdnbbsr.s3waas.gov.in/s3716e1b8c6cd17b771da77391355749f3/uploads/2023/10/20231005118040475.pdf

[25] Source: https://powermin.gov.in/sites/default/files/webform/notices/Notification_Regarding_Renewable_Purchase_Obligation_RPO.pdf

[26] Note: While the Ministry of Power, Govt. of India has provided RPO targets till FY30, an escalation assumption of increase in RPO target between FY29 and FY30 has been assumed to continue till FY32 for the purpose of this analysis

S. No	Particulars	Unit	FY23	FY24	FY25	FY26	FY27	FY28	FY29	FY30	FY31	FY32
2	Wind RPO	GWh	100	210	94	216	312	413	529	664	814	982
3	Hydro RPO	GWh	43	87	53	181	212	239	255	254	252	249
4	Distributed Energy Resource RPO	GWh	-	-	105	156	214	278	350	429	518	616
5	Other RPO	GWh	2,890	3,256	3,925	4,355	4,952	5,607	6,284	6,921	7,615	8,369
6	**Total RPO**	**GWh**	3,033	3,553	4,177	4,908	5,689	6,537	7,417	8,269	9,199	10,215

In terms of Renewable Energy (RE) potential, as per Central Government estimates, Assam has a potential of around 14.24 GW of possible RE implementation in state. Out of this, solar potential makes up for 13.76 GW followed by small hydro and bio-power at 202 MW and 279 MW respectively. The State has no potential for Wind Energy and has been resorting to buy wind power from outside state. Even for small hydro and biopower the state has historically faced challenges with respect to geographic terrain, gestation period and supply chain issues respectively. **Thus, to meet the RPO targets, specifically for the Distributed Energy Resource (DER) and Other Category, Assam will have to rely heavily on solar power.**

6.3 Planned Capacity Addition of Generation Projects

The Government of Assam's Renewable Energy (RE) Policy for 2022 aims to add 1200 MW (1000 MW from solar and 200 MW from other sources) of renewable energy by 2027. Additionally, the government plans to install another 1000 MW of solar power on unused state government land. To align with this policy and address Assam's growing electricity demand, the Assam Power Generation Company Limited (APGCL) and the Assam Power Distribution Company Limited (APDCL) aim to increase capacity by approximately 4.2 GW by the fiscal year 2030. The proposed projects include:

Table 5: Capacity Addition Projects Pipeline in Assam

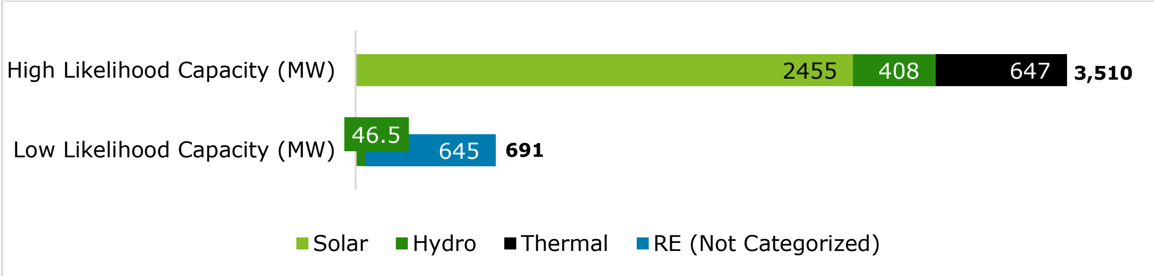
S No	Name of project	Capacity (MW)	Type	Agency	Expected COD (Financial Year)	Remarks	Likelihood of Planned Capacity to materialize considering progress and timelines[27]	Capacity with high likelihood to be commissioned as per timeline	Capacity with Low likelihood to be commissioned as per timeline
1	3 X 660 MW NUPPL project	493	Thermal	APDCL	2024	Power Purchase Agreement (PPA) signed by APDCL with the project	High	493	-
2	NHPC Subansiri	208	Hydro	APDCL (Allocation)	2024	Allocation from the Government of India; Project Construction Underway	High	208	-
3	Nikachhu Hydro Electric Project	80	Hydro	APDCL (Import from Bhutan)	2024	PPA signed by APDCL and progressing well	High	80	-
4	10 MW APDCL solar project under CPSU Ph-II scheme	10	Solar	APDCL	2024	Tender Process Underway	High	10	-
5	Namrup Solar PV Project	25	Solar	APGCL	2025	Detailed Project Report Finalized	High	25	-
6	Lower Kopili HEP	120	Hydro	APGCL	2025	Construction underway	High	120	-
7	APDCL Solar Project (ADB)	500	Solar	APDCL	2026	Proposed 1000 MWp project (ADB)	High	500	-
		500	Solar		2027			500	-
8	50 MW APDCL solar project on PPP basis	50	Solar	APDCL	2026	Bid Process Underway	High	50	-
9	70 MW APDCL solar project on PPP basis	70	Solar	APDCL	2026	Bid Process Underway	High	70	-
10	200 MW APDCL solar project on PPP basis	200	Solar	APDCL	2026	Bid Process Underway	High	200	-
11	RE Projects - APGCL	322.5		APGCL	2026		Low	-	322.5

[27] Firm Capacity has been considered basis on-ground progress details available.

S No	Name of project	Capacity (MW)	Type	Agency	Expected COD (Financial Year)	Remarks	Likelihood of Planned Capacity to materialize considering progress and timelines[27]	Capacity with high likelihood to be commissioned as per timeline	Capacity with Low likelihood to be commissioned as per timeline
		322.5	Renewable Energy	APGCL	2027	Proposed by APGCL; No further details available	Low	-	322.5
12	NTPC Talcher III	154	Thermal	APDCL (Allocation)	2027	Central Govt. Allocation for Assam; Project under progress as per CEA	High	154	-
13	APDCL Solar (JV with NLCIL)	250	Solar	APDCL	2026	Memorandum of Understanding (MOU) signed between APDCL and NLCIL	High	250	-
		250	Solar		2027			250	-
		250	Solar		2028			250	-
		250	Solar		2029			250	-
14	Small Hydro Projects - APGCL	46.5	Hydro	APGCL	2028	Proposed by APGCL and project details are not available; Hydro projects in North-Eastern India usually take minimum 5 years to construct. Since this project has not yet begun construction, it is assumed that it may not come-up within the expected COD	Low	-	46.5
15	100 MW APDCL floating solar project on PPP basis	100	Solar	APDCL	2030	Proposed by APDCL and since expected COD is 2030 and it is a solar project, it may come-up within the time-frame	High	100	-
Total		**4,201 (MW)**						**3,510 (MW)**	**691 (MW)**

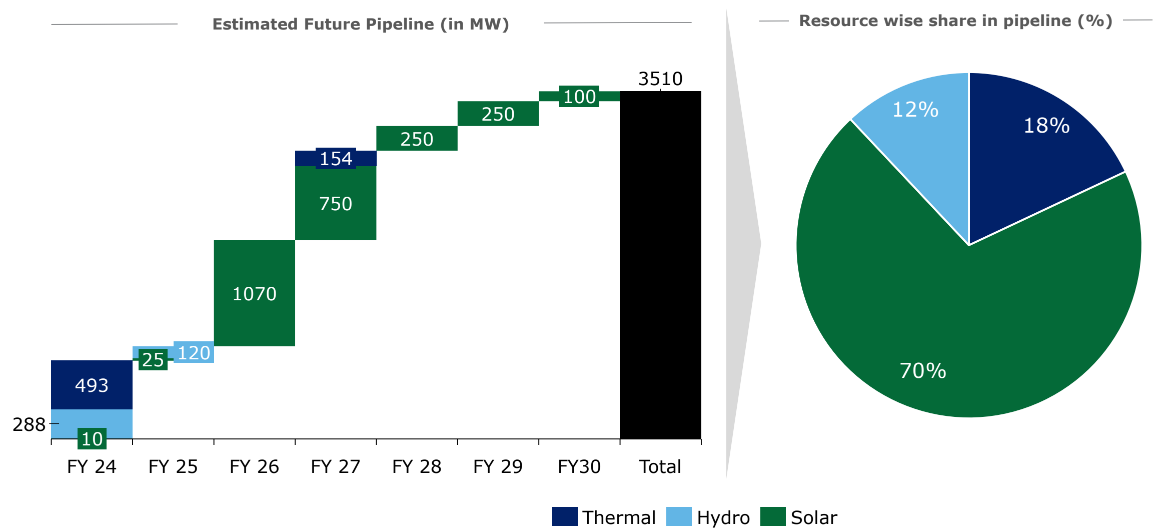
Apart from the above project pipeline, APGCL also aspires to add 1.2 GW of additional capacity by FY30. However, the same is not being considered firm as no on-ground progress is reflected for the projects. Out of the planned 4.2 GW of capacity, ~3.5 GW has a high likelihood to be commissioned as per timelines based on the progress of the project. The remaining ~0.7 GW MW has a low likelihood of being commissioned as per the timelines and may require a re-look in terms of project planning and progress. Thus, the following is the breakup of the capacity in terms of high and low likelihood which may be commissioned by FY2030.

Figure 24: Capacity with High and Low Likelihood expected to come up in Assam by FY2030



In formulating the assessment, emphasis has been placed on determining the solar capacity requirement for Assam by exclusively considering projects with a high likelihood of execution within the FY30 timeline. Out of the 3.5 GW capacity deemed highly probable, approximately 70%, equivalent to around 2.5 GW solar based capacity, while thermal and hydro sources contribute approximately 0.6 GW and 0.4 GW, respectively. This approach ensures a realistic plan for solar power capacity addition by focusing on projects that are very likely to come online.

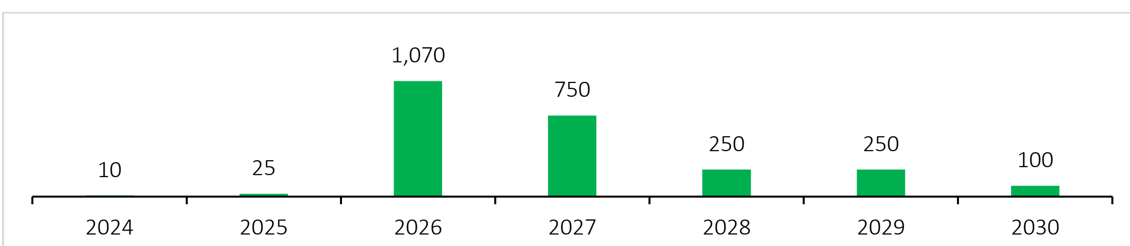
Figure 25: Estimated Future Pipeline (in MW) – Firm Capacity as per on-ground progress



6.4 Solar Power Capacity Addition Requirement

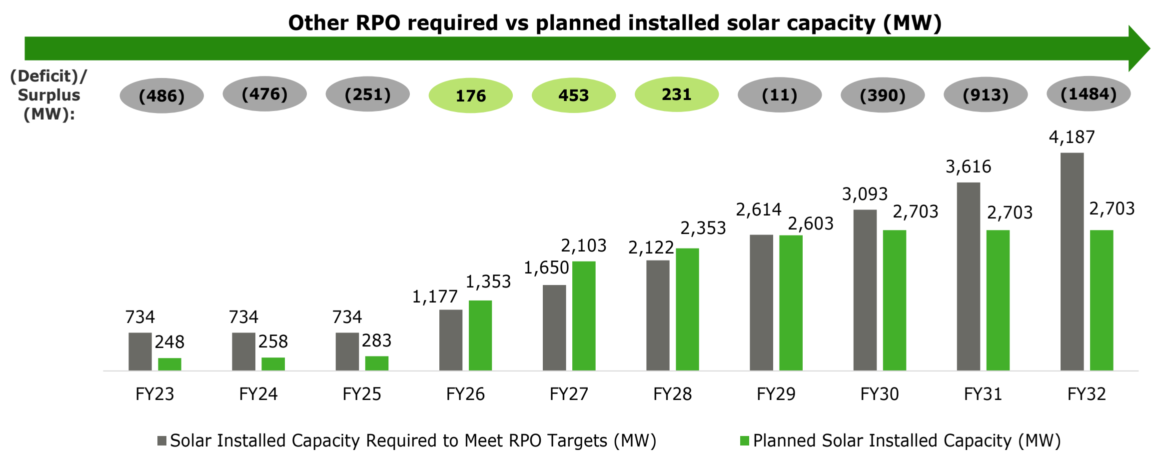
The Year-on-Year (YoY) Solar Capacity Addition (High Likelihood) in the state based on the above project plan has been showcased below:

Figure 26: YoY Solar Capacity Addition Planned (MW) – High Likelihood – Total 2,455 MW



Even with the proposed capacity addition of 2.5 GW of solar and other hydro and thermal capacity in the high likelihood scenario, the RPO targets for Assam would require the state to have ~4.2 GW of Solar Capacity of FY2032[28]. This is considering that the targets for DER and Other RPO categories are met by solar power based on the solar potential for the state. However, it is to be noted that the category wise energy requirement for various types of RPO may not be arithmetically same as the percentage (%) targets for each year. This is due to the guidelines allowing for any surplus/shortfall in any category to be met through other renewable energy sources. For e.g., any shortfall in achievement of stipulated wind renewable energy consumption in a particular year may be met with hydro renewable energy which is more than that energy component for that year and vice versa. The energy requirement targets in the table below considers the planned portfolio of all type of projects in the state.

Figure 27: Solar Installed Required to meet RPO Targets vis-a-vis Planned Solar Capacity Addition in Assam (High Likelihood Scenario)



Key Conclusion:

1. **Capacity Addition Shortfall in Early Years (FY23-FY25):** In the initial years (FY23 to FY25), there is a consistent shortfall in planned solar installed capacity compared to the required capacity. The shortfalls range from 251 MW to 486 MW, indicating the need for accelerated efforts to meet the solar capacity targets.

2. **Surplus in Mid-Years (FY26-FY28):** Between FY26 to FY28, there is a surplus in planned solar installed capacity compared to the required capacity. This surplus ranges from 176 MW to 453 MW, showcasing successful efforts in exceeding the solar capacity targets during this period.

3. **Large Capacity Addition Requirement in Later Years (FY29-FY32):** FY29 and FY30 experience small shortfalls, while FY31 and FY32 show substantial shortfalls, emphasizing the need for strategic adjustments to align with long-term targets. There is a need for significant capacity addition of 2.8 GW additional solar capacity (over the planned capacity) to bridge the widening shortfalls.

4. **Long-Term Vision (till FY32):** The roadmap highlights the journey from an initial shortfall to achieving surpluses, followed by renewed challenges. A holistic, long-term commitment is essential for sustained success, requiring proactive measures to address emerging trends, technology advancements, policy changes and private sector participation.

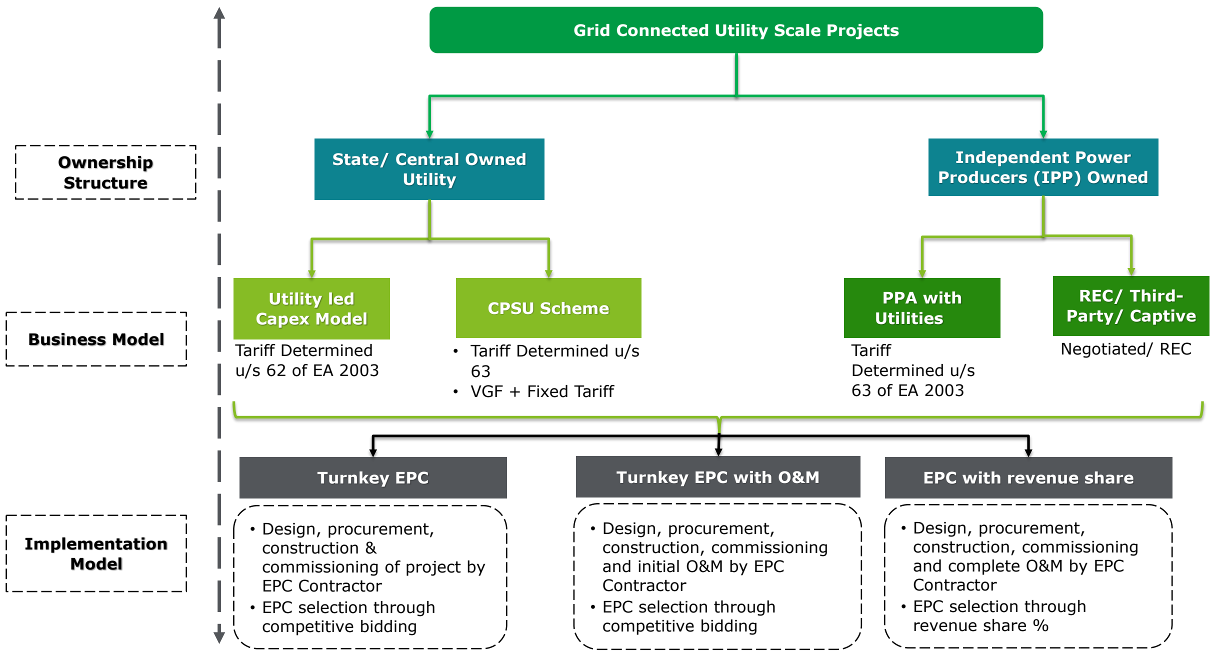
[28] Source: Deloitte Analysis; Assumed Other + DRE RPO to be met by Solar Capacity at average CUF of ~19% (as per AERC Norms for CUF for Solar Projects under RE Tariff Regulations, 2017); Other RPO Trajectory for FY31 and FY32 extrapolated based on incremental RPO between FY29 and FY30 as set by MoP/AERC

7 Business Models for Solar Projects and Learnings from Project Development Experience in Assam

In this section, prevalent business models for setting up Solar PV projects has been assessed which provides a foundation for developing Solar Projects in Assam going forward.

In terms of Grid Connected Utility Scale Solar Projects, two (2) types of ownership-based structures are prevalent i.e. Utility Owned and Independent Power Producer (IPP) Owned. The below graph illustrates the ownership structure wise business and implementation models possible under each structure:

Figure 28: Business models for utility scale solar projects in India

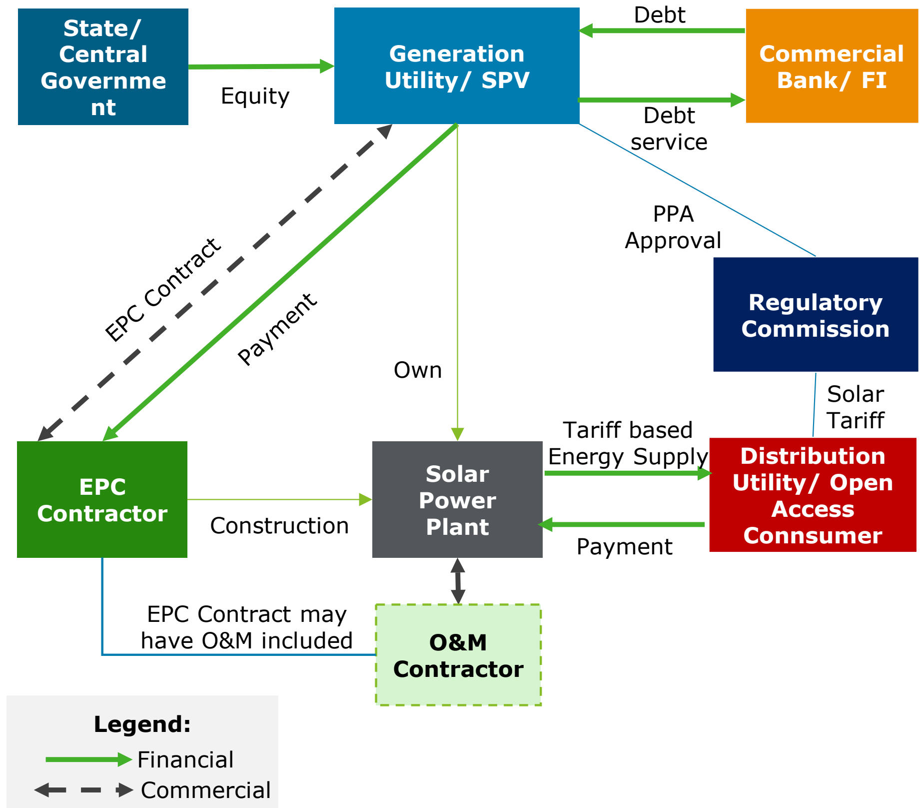


Among the above models, IPP owned projects governed by PPAs with utilities hold the maximum presence in the country, wherein tariffs are discovered competitively u/s 63 of Electricity Act 2003.

7.1 Public Sector owned utility scale solar Model

This is a utility led model wherein capex is incurred by the utility itself; thus, ownership of such projects also lie with the utilities. Typically, in such scenarios, a Turnkey/ package-wise EPC contract is/are awarded to Vendor(s) through competitive bidding/ nomination basis.

Figure 29: Utility led Capex Model



Key Highlights of the model:

Table 6: Key Highlights of Utility Owned Capex Model

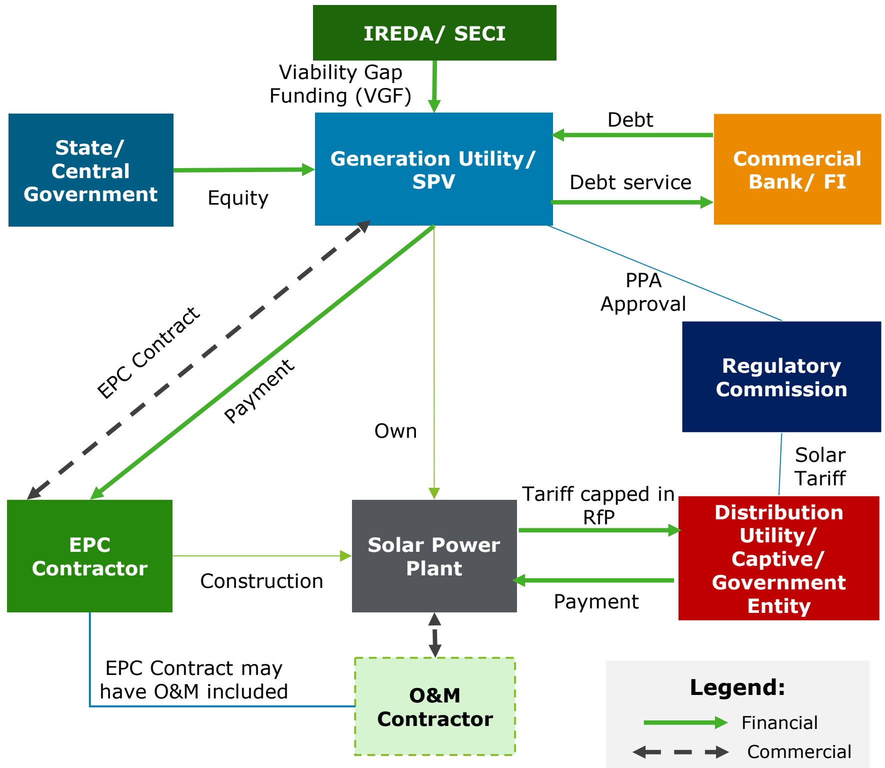
Aspect	Details
Key Features	• Government Utility builds, owns and operates the solar project • Suitable for small to large scale projects
Financing	• Equity funded by State Government/ Utility – Balance sheet based funding • Project specific Debt raised from commercial banks/ FIs
Contractual Arrangement	• Power Purchase Agreement – Where generation utility and procuring utility are different or SPV is used for solar project. No PPA is required where procuring and generating utilities are same • EPC Agreement – Between the generating utility and EPC contractor
Commercial	• Regulator to determine the tariff on cost plus (U/S 63) • Competitive selection of EPC (adoption of tariff U/S 62)
Pros and Cons	**Benefits** • Less complexity in transaction structuring • Builds expertise of Utility for future projects • Greater control in cost optimization **Challenges** • Utility's ability to develop solar projects and bring efficiencies • Limited experience in operating and managing solar projects

In terms of the utility model, 2 case studies of NTPC owned projects have been enumerated in Annexure 9.7

7.2 CPSU Scheme for solar projects

This scheme gives opportunity for Central and State PSUs (CPSU) to execute solar projects with Viability Gap Funding (VGF). Under CPSU scheme, Government/ Government entities can participate in tenders wherein tariff would be capped, and basis of bid participation would be quantum of Viability Gap Funding (VGF) quoted. Responsibility of finding buyers lies with the Bid winners. The transactional overview as well key highlights of the model have been enumerated below:

Figure 30: CPSU Model - Project Overview



Key Highlights:

Table 7: Key Highlights of the CPSU Scheme

Aspect	Details
Key Features	• Ministry of New & Renewable Energy (MNRE) approved implementation of CPSU Scheme Phase-II for setting up 12,000 MW grid-connected solar projects by CPSUs/ State PSUs/ Government Organizations, with Viability Gap Funding (VGF) support
Financing	• Equity funded by State Government/ Utility – Balance sheet-based funding • Project specific Debt raised from commercial banks/ FIs
Contractual Arrangement	• For self-use or use by Government/ Government entities, either directly or through Distribution Companies (DISCOMS).
Commercial	• VGF capped at Rs 0.55 Cr /MW - decided through bidding • VGF released in two tranches - 50% on EPC contract Award & balance on commissioning of full capacity • Tariff capped at Rs 2.45 per kWh
Pros and Cons	**Benefits** • Less complexity in transaction structuring • Builds expertise of Utility for future projects **Challenges** • Can lead to inefficiencies in procurement • Limited experience in operating and managing solar projects

The results of some of the tenders under CPSU Scheme have been illustrated below:

Figure 31: Results of tenders under CPSU Scheme[29]

SECI TRANCHE I (Aug 2019)
- Targeted Capacity : 2 GW
- Awarded Capacity: 922.4 MW
- Capped Tariff: Rs 3.50 per kWh

Bidder's Name	VGF Quoted (Rs L/MW)	Capacity (MW)
NHDC Limited	55.00	25
Singreni Collieries Company	60.00	90
Assam Power Distribution (APDCL)	68.00	30
Delhi Metro Rail Corporation (DMRC)	69.75	3
Nalanda University	69.96	5
NTPC Limited	70.00	769

SECI TRANCHE II (Nov 2019)
- Target Capacity: 1.5 GW
- Awarded Capacity: 1104.8 MW
- Capped Tariff: Rs 3.50 per kWh

Bidder's Name	VGF Quoted (Rs L/MW)	Capacity (MW)
Singreni Collieries Company	68.00	81
Indore Municipal Corporation	68.80	100
NTPC Limited	70.00	923

IREDA TRANCHE III (Sep 2021)
- Target Capacity: 5 GW
- Awarded Capacity: 5 GW
- Capped Tariff: Rs 3.50 per kWh

Bidder's Name	VGF Quoted (Rs L/MW)	Capacity (MW)
SJVN Limited	44.72	1,000
NLC India Limited	44.75	510
NHPC Limited	44.90	1,000
IRCON International Ltd	44.94	500
NTPC Limited	44.95	1,990

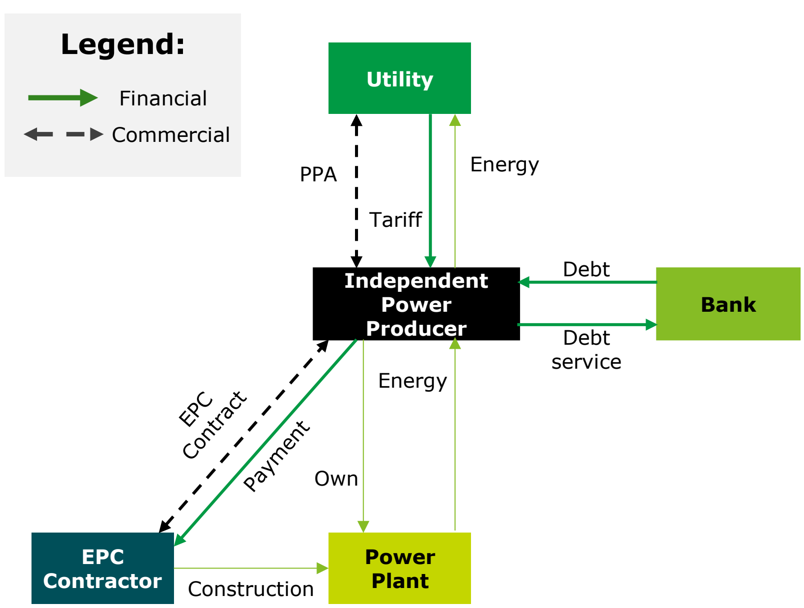
7.3 Private Owned utility scale solar models

The projects under IPP / PPP based ownership are financed by Private Independent Power Producers (IPPs) or combination of IPP along with Public utility. Ownership of projects is dependent on the financing structure. Generally, tenders are issued by Govt. utilities for which reverse bidding takes place for transparent discovery of solar tariff under Section 63 of Electricity Act 2003. Plain vanilla solar tenders are typically of two types predominantly, viz. solar park (BOOT) and Non-Solar Park (BOO models).

7.3.1 Build Own Operate (BOO) Model

In the BOO Model, the project land and complete responsibility of connectivity inter alia lies with the Project Developer. Complete ownership of project by IPP Developer / PPP Consortium.

Figure 32: Project Overview for BOO Model



[29] Source: Deloitte Research

Key Highlights:

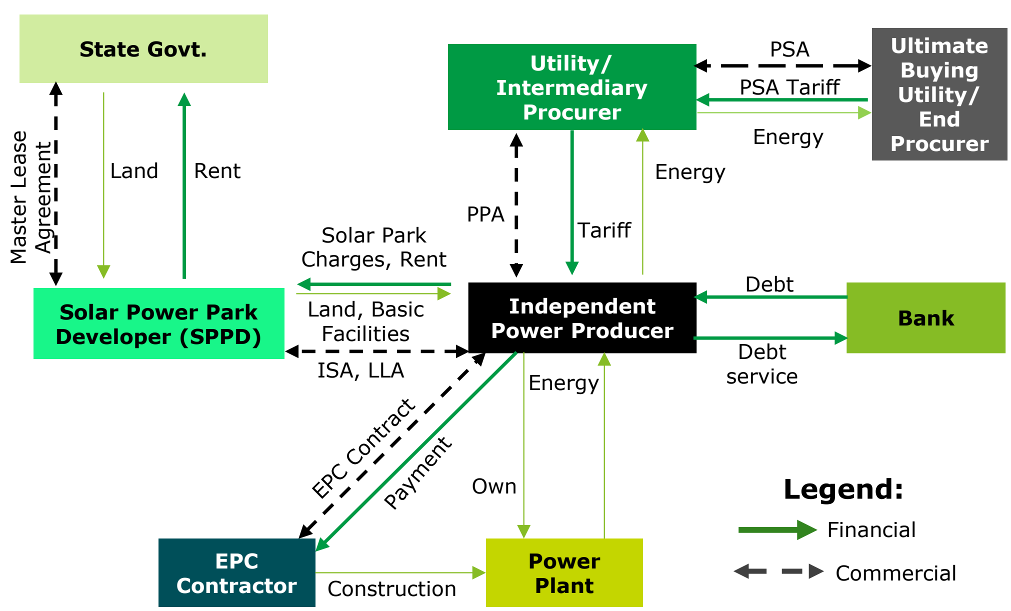
Aspect	Details
Key Features	• Suitable for large grid connected solar projects – can be within a solar park or independently developed • Based on BOOM or BOOT Model
Financing	• Equity by Private developers • Project specific debt from commercial banks/ FIs • Solar Park (wherever applicable) developed through FI funding + Government equity
Contractual Arrangement	• PPA signed with state distribution utility/ open access consumer or third party • ISA/ land lease agreement with solar park developer (if applicable) • Transmission Service Agreement with the transmission utility (State/ Central)
Commercial	• Tariff based competitive bidding (TBCB) for procurement by utilities/ government agencies
Pros and Cons	**Benefits** • Cost optimization through TBCB route • Participation of Private players with niche expertise **Challenges** • Delays in case of dependence on other agencies for clearances

Details of BOO based solar projects implemented under SECI scheme is provided as Annexure 9.5

7.3.2 Build Own Operate Transfer (BOOT) – Solar Park Model

In case of solar park tenders, developers establish solar projects in pre-identified lands with clear demarcations; wherein responsibility of land and connectivity lies with Solar Park Implementing Agency (Public / JV with State Govt.) (SPIA). SPIA signs Master Lease Agreement with State Govt. and in turn signs Sub-Lease Deeds with Developer as well as Implementation Support Agreement (ISA). The Developer signs PPA with Intermediary Procurer/ Procurer and Intermediary Procurer in turn signs PSA with Ultimate Buying Entity. Since land is govt. owned, post PPA term, assets are transferred by Developer to Govt. entity/PPA party.

Figure 33: Project Overview of BOOT model (Solar Park)



Key Highlights:

Aspect	Details
Key Features	• Land, solar park infrastructure along with off take infrastructure from PSS to GSS provided by Solar Park Implementing Agency (Public/JV) • Project Construction, connectivity upto PSS and O&M by Private Developer • After 25 years, asset transferred from Private to SPIA (Public/ JV)
Pros	• Ready land and basic infrastructure at the outset • Shorter turnaround time for project execution
Cons	• Implementing the model can be complex as it involves numerous transactions

State-wise details of solar parks as on Dec 2022 as Annexure **9.6**

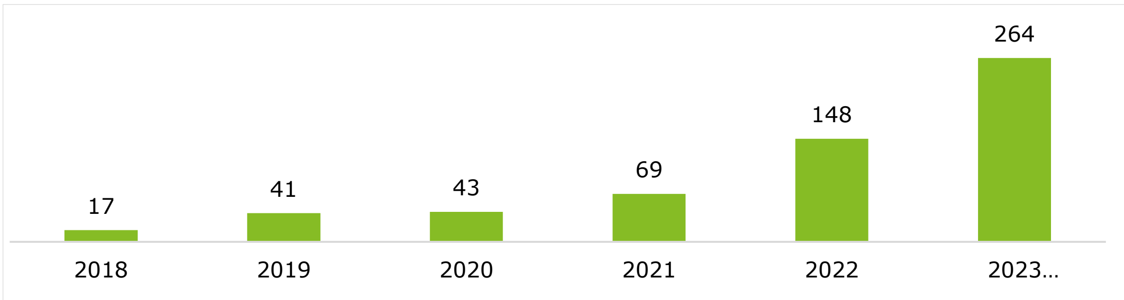
Each business model presents unique advantages and challenges. The Public Sector Owned Utility Scale Solar Model could be beneficial for building local expertise, but the government's limited experience in managing large solar projects may pose challenges. The CPSU Scheme simplifies transaction structuring but requires careful consideration towards achieving the benchmarked tariffs. Private-owned models (BOO and BOOT) offer cost optimization and attract private players, but they may face delays and complexities in certain situations. Assam needs to carefully evaluate these models based on its specific requirements, resource availability, and administrative capabilities to choose the most suitable approach for sustainable solar project implementation. To identify suitable business model for implementation or transition between one model to another, it is also important to identify the key challenges and learnings from solar project development in the state.

7.4 Learnings from Solar Project Development Experience in Assam

7.4.1 Current Status

In the last 5 years, Assam has added ~223 MW of solar capacity increasing the installed base by ~5X times. Most of the projects have been small-scale IPP projects with the largest single location plant having a capacity of 70 MW at Amguri. The discovered tariff for these projects has been in the range of INR 3.2-4 per kWh.

Figure 34: YoY Solar Installed Capacity Trend (MW)



In terms of the project locations, the concentration has been largely within Central Assam where most of the demand centers also currently fall. The location wise small scale solar projects have been illustrated below:

Figure 35: Solar utility scale projects located in Assam

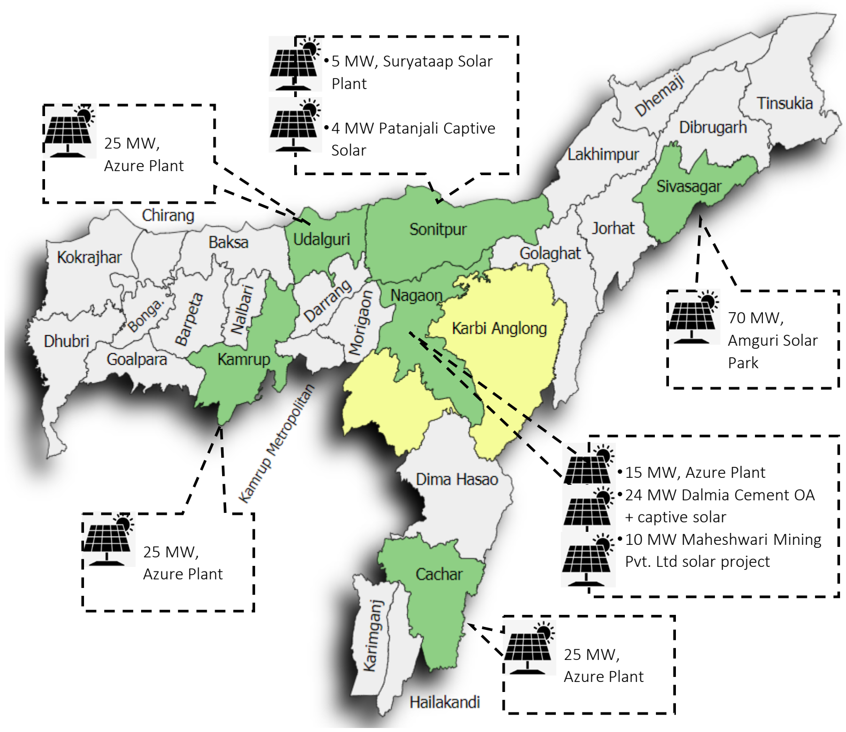


Table 8: List of utility scale solar projects in Assam

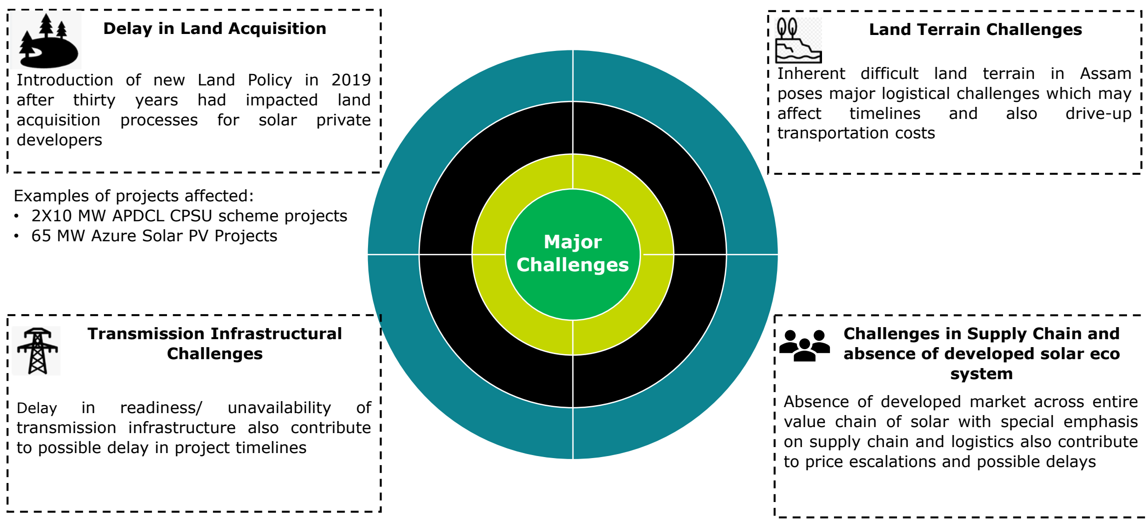
S No	Implementing Agency	Contract	Capacity (MW)	COD	Model
1	Suryataap Solar	PPA	5	20-Aug-16	-
2	Azure Power	PPA	25	12-Sep-20	BOO

S No	Implementing Agency	Contract	Capacity (MW)	COD	Model
3	Azure Power	PPA	25	30-Dec-21	BOO
4	Azure Power APDCL	PPA	15	27-Jan-22	BOO
5	Azure Power	PPA	25	31-Mar-22	BOO
6	Maheshwari Mining & Energy	PPA	10	31-Mar-22	BOO
7	APGCL (Dev.: Jakson)	Solar Park	70	31-Mar-22	BOOT
8	Patanjali	Captive	4	NA	-
9	Dalmia Cement	Captive + OA	24	NA	-
		Total	**203**		

7.4.2 Key Learnings

While solar capacity has risen in Assam, the projects have historically faced multiple challenges pertaining to land acquisition, difficult terrain, power evacuation as well as solar supply chain and ecosystem. Some of these challenges have been enumerated below:

Figure 36: Key Challenges in setting up of Solar Projects in Assam



In order to scale the State's Solar Capacity to ~4200 MW by FY32, there is a need to plan mitigation techniques through policy, regulatory measures and interventions to support scalable business models. The next section discusses the way forward for the project development and applicability of various business models in the state.

8 Way Forward

The State of Assam has witnessed a Compound Annual Growth Rate (CAGR) of ~10% in aggregate State GDP in the last 5 years (FY17 to FY22)1 which is more than the National and NER growth rates. The State is a focus area state for the Govt. of India due to its strategic location and access point to the rest of NER. Backed by the economic growth, the power demand in Assam has been increasing at a CAGR of ~7.8% since FY162 and is expected to grow at a CAGR of 6.4% during the next decade till FY32.

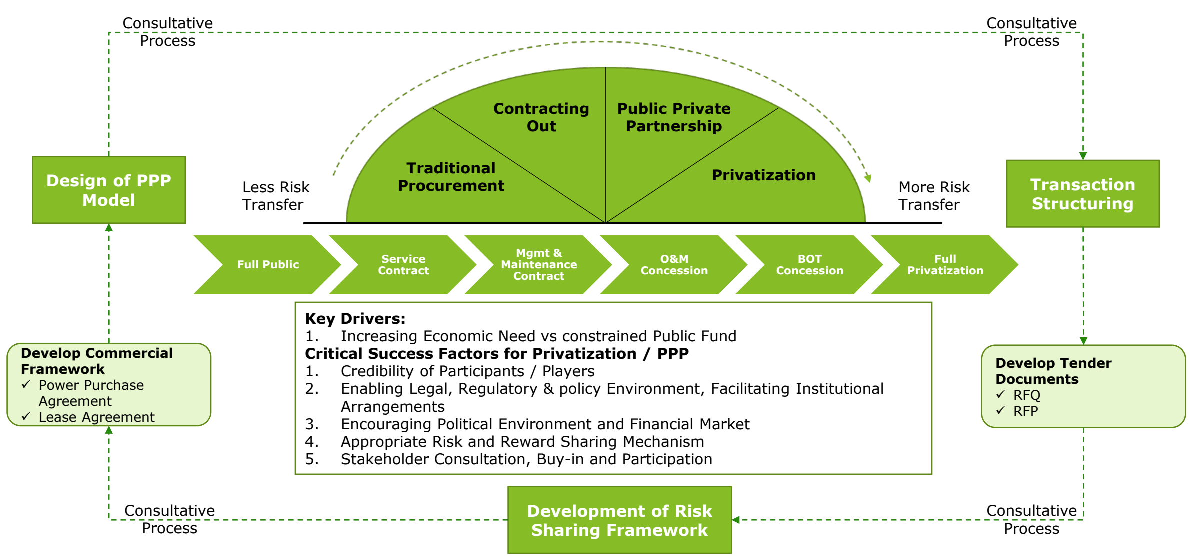
Assam's own electricity supply portfolio which has historically been dominated by Gas and Hydro based power plants are facing challenges ranging from wide price fluctuations, availability of gas as well as environment and gestation challenges. The State's APPC of INR ~5.4 per kWh in FY23 makes it one of the highest in the region.

The current energy landscape in Assam faces significant challenges, especially with the introduction of the new Renewable Power Obligation (RPO) mandated by the Ministry of Power (MoP). According to the proposed trajectory, Assam must strive for approximately 4200 MW of solar capacity by 2032 to fulfill its RPO. Despite possessing substantial potential for solar projects, the state is resorting to purchasing green power from short-term markets, even during solar hours, due to rising demand and inter-state transmission capacity constraints.

Given the limited resource potential for coal, wind, hydro, small hydro, and biomass in the state, solar emerges as the primary resource with a significant capacity of 13.7 GW for power supply expansion. However, Assam has faced challenges in implementing Independent Power Producer (IPP) and Public-Private Partnership (PPP) projects. Issues such as difficult terrain, high transportation costs, and power evacuation challenges have deterred private players, resulting in the realization of only a modest 70 MW capacity at a single location.

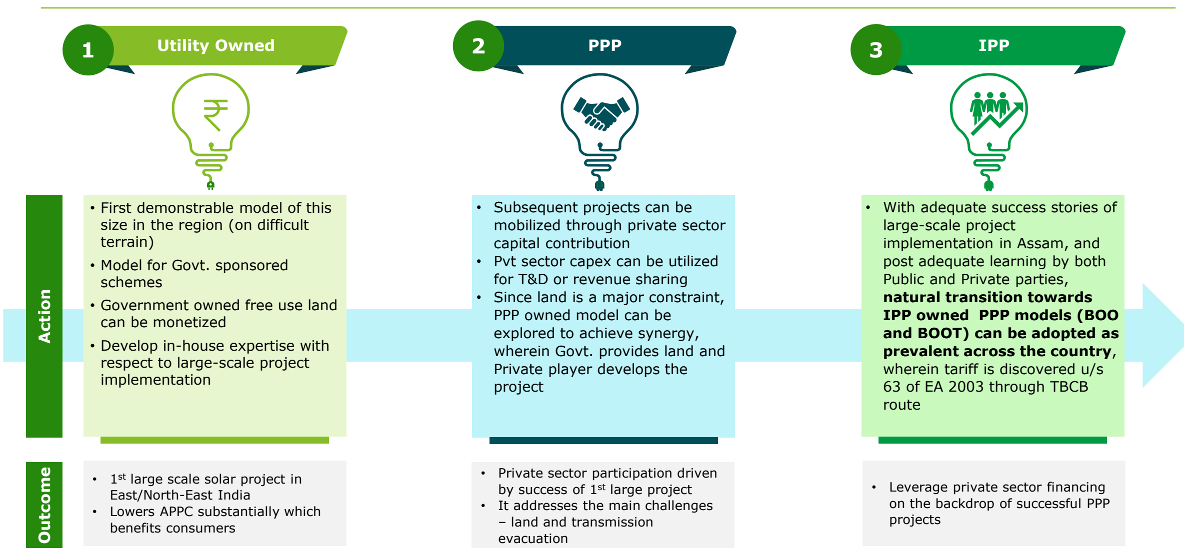
In response to these challenges, it becomes imperative to expedite the establishment of renewable energy projects in Assam, particularly at a large scale, to meet the surging demand and fulfill RPO requirements. The state utilities' support is crucial in initiating successful demonstrable large-scale projects. While private participation can be incorporated, a utility-led capital expenditure (capex) model with initial public utility responsibilities is envisioned. This strategic approach aims to address existing implementation challenges, such as land and evacuation issues, and lay the foundation for a gradual transition to PPP and eventually IPP models in the state.

Figure 37: Development of Implementation models for solar in a developing market



The phased approach, beginning with a utility-led capex model, ensures that the state utilities play a pivotal role in the early stages, taking on significant responsibilities. This involvement not only helps overcome initial hurdles but also establishes a framework for future large-scale projects and the achievement of market maturity in the state. As major capacity projects under the utility-led model prove successful, private participation can be gradually increased, leading to a natural evolution towards more private sector-led models. This transition allows for the gradual transfer of risks from the public to private players, fostering a sustainable and progressive development trajectory for renewable energy projects in Assam. Overall, this strategic and phased approach aims to create a conducive environment for the successful execution of large-scale projects, paving the way for Assam to meet its renewable energy targets and contribute substantially to the country's clean energy goals.

Figure 38: Impact of Various Implementation Models



9 Annexure

9.1 Institutional Setup and Financial Performance of the Public Entities in Assam Power Sector

9.1.1 Assam Electricity Regulatory Commission (AERC)

The Assam Electricity Regulatory Commission (hereinafter referred to as the AERC or the Commission) was established under the Electricity Regulatory Commissions Act, 1998 (14 of 1998) on February 28, 2001. The first proviso of Section 82(1) of the Electricity Act, 2003 has ensured continuity of the Commission under the Electricity Act, 2003. Major function of AERC has been regulating the electricity sector in the State in accordance with the provisions of the Electricity Act 2003. The AERC has put in place the key regulations governing the sector and has been issuing tariff orders for the utilities.

With respect to RE implementation in state AERC plays a crucial role wherein it is responsible for inter alia, issuance of policies facilitating Renewable Energy (RE) implementation, setting up of RPO targets/ aligning with national targets coupled with monitoring of actual achievement, adoption of tariff under Section 63 for (RE) tariff discovered under Tariff based Competitive Bidding (TBCB) or under Section 62 for Cost plus model under Electricity Act (EA) 2003; PSA/ capacity approval under Section 86 (1) (b) of EA 2003; promoting generation of electricity from renewable sources of energy by providing suitable measures for connectivity with the grid as per Section 86 (1) (e) of the Electricity Act. Thus the regulatory commission in state plays a key part in ensuring that its respective state is conducive for successful implementation of power projects with special emphasis to RE implementation.

9.1.2 Assam Power Generation Corporation Ltd. (APGCL)

APGCL is the successor corporate entity of erstwhile Assam State Electricity Board (ASEB) formed pursuant to the notification of the Government of Assam, notified under sub-sections (1), (2), (5), (6) and (7) of Section 131 and Section 133 of the EA 2003 (Central Act 36 of 2003), for the purpose of transfer and vesting of functions, properties, interests, rights, obligations and liabilities, along with the transfer of personnel of the Board to successor entries. APGCL is a Company incorporated with the main object of generation of electricity in the State of Assam and is a Generating Company under the various provisions of the Act.

The certificate of commencement of business was obtained w.e.f. 29th April 2004. The final Transfer scheme was implemented in August 2005 with a new Company Balance Sheet w.e.f. April 2005.

The company is mainly responsible for:

- Maximum energy generation to meet up the energy demand in the state.
- Operation & Maintenance of all existing Power Generation Plants & electrical system infrastructure associated with evacuation of power within the power station.
- Completion of on-going Power Projects so approved by Govt. of Assam (GoA) and
- Arrangement of necessary power evacuation in co-ordination with Assam Electricity Grid Corporation Ltd.

Further, with respect to RE implementation in state, APGCL, the state power Generation Company (Genco) will play a significantly important role with respect to energy transition in state and in the process possibly reducing dependency on gas-based generation wherein the prices and availability of the said resource is extremely volatile. With respect to the same, APGCL has already made some mentionable strides through aiding the development of 70 MW Amguri Solar project. Further, currently there is approximately 836 MW RE projects in pipeline (solar and hydro). Successful implementation of above in timely manner would be significant boost to the RE portfolio in state. Their role would be crucial to increase reliability on state's own generation to achieve their required RPO.

Financial Snapshot of APGCL[30]:

Table 9:Financial Snapshot - APGCL

Amount figures in INR Crores

Item	FY2018	FY2019	FY2020	FY2021	FY2022
Commercial Indicators					
Electricity transmitted grand total	1,404	1,510	1,506	1,330	1,996
Average revenue per unit (₹/kWh)	3.65	3.89	3.72	3.31	2.68
Average cost per unit (₹/kWh)	2.66	2.81	2.79	2.46	2.84
Financial Indicators (₹ Crores)					
Total Revenue	512	587	559	440	535
Total Expenses	486	555	556	467	680
Expenses (excluding Depreciation and Finance Cost)	373	424	420	327	568
Operating profit	139	163	139	114	(33)
Interest expenses	73	79	80	84	27
Gross Fixed Asset	1,224	1,559	1,572	1,606	2,516
Net Fixed Asset	562	844	803	780	1,606
Current Asset	1,117	1,249	1,205	1,133	1,126
Current Liability	603	948	1,052	1,152	311
Non-Current Liability	1,089	842	802	709	552
Total Equity	1,237	1,355	1,399	1,667	2,555
Depreciation and amortization	40	53	56	56	85
Exceptional Items	-	-	-	-	396
Gross profit	26	32	3	(26)	251
Tax	6	7	1	-	43
Net Profit After Tax	20	24	3	(26)	208
Profit Ratios					
Operating profit margin	28.61	29.35	25.02	24.38	(4.79)
Net profit margin	3.99	4.14	0.46	(5.99)	38.84
Leverage Ratio (D/E)	1.37	1.32	1.33	1.12	0.34
Debt to Capital Ratio	0.58	0.57	0.57	0.53	0.25
Return on Net Assets	0.02	0.02	0.00	(0.03)	0.09

[30] APGCL Annual Accounts

In FY 22, increase in PAT despite higher expense has been due to influx of Exceptional item of amount Rs 396.48 Cr, which in turn is due to Govt. of Assam's waiver of interest to APGCL thus becoming income of APGCL.

9.1.3 Assam Electricity Grid Corporation Ltd (AEGCL)

AEGCL is the successor corporate entity of erstwhile Assam State Electricity Board (ASEB) formed pursuant to the notification of the Government of Assam (GoA), notified under subsections (1), (2), (5), (6) and (7) of Section 131 and Section 133 of the EA 2003 (Central Act 36 of 2003), for the purpose of transfer and vesting of functions, properties, interests, rights, obligations and liabilities, along with the transfer of personnel of the ASEB to successor entries. AEGCL owns and operates the transmission system previously owned by ASEB. AEGCL has started functioning as a separate entity from December 10, 2004.

Intra-state transmission is carried out by AEGCL. Transmission infrastructure has not witnessed significant capacity addition in past few years as the existing capacity has been able to serve the peak demand in the State. The transmission line in the 220 KV increased by 324 Kms between FY19 and FY20. No other major line extension has taken place during FY18 and FY22. In terms of the Transformer Capacities, AEGCL added 824 MVA (Extra High Voltage) at a CAGR of 4% between FY20 and FY23 (till August 2022); and with respect to grid substations, AEGCL added about 5 substations (net) between FY18 and FY22 of which the majority has been added in the 132 / 33 kV voltage level (4 in total)[31].

However, as the State continues to witness rapid demand growth, the same may need to be augmented to ensure reliable supply and expand to newer demand centers in the State. Further, significant capacity addition in terms of power projects is envisaged in the state of Assam in the next decade as has been detailed in the subsequent section. A significant part of this capacity addition would be through RE projects. And thus, adequate grid infrastructure and evacuation capacity would be instrumental to ensure that the plants can operate optimally. Further, with respect to AEGCL would also have to ensure timely grant of connectivity to projects seeking connectivity to STU to ensure there are no delays on account of delay in grant of connectivity.

Financial Snapshot of AEGCL[32]:

Table 10: Financial Snapshot - AEGCL

Amount figures in INR Crores

Item	FY2018	FY2019	FY2020	FY2021	FY2022
Commercial Indicators					
Total Energy Sent Out (GWh)	8,605	8,930	9,033	9,482	10,529
Average revenue per unit (₹/kWh)	1.63	1.45	0.38	0.48	0.54
Average cost per unit (₹/kWh)	0.99	1.13	0.40	0.30	0.32
Financial Indicators (₹ Crores)					
Total Revenue	1,401	1,296	343	452	568
Total Expenses	949	1,124	525	475	463

[31] Source: Petition for True Up for FY 2021-22, Annual Performance Review for FY 2022-23 and Aggregate Revenue Requirement for FY 2023-2024 of AEGCL

[32] Source: AEGCL Annual Accounts

Item	FY2018	FY2019	FY2020	FY2021	FY2022
Expenses (excluding Depreciation and Finance Cost)	848	1,009	365	286	337
Operating profit	553	288	(23)	166	232
Interest expenses	41	50	58	73	2
Gross Fixed Asset	1,682	1,973	2,394	2,563	2,738
Net Fixed Asset	606	825	1,140	1,187	1,222
Current Asset	1,354	1,666	1,468	1,576	1,709
Current Liability	1,030	1,120	1,128	1,548	1,083
Non-Current Liability	678	674	687	288	681
Total Equity	1,485	1,829	1,717	1,784	2,126
Depreciation and amortization	60	65	102	116	124
Exceptional Items	-	-	-	-	-
Gross profit	452	173	(182)	(23)	105
Tax	95	27	-	-	-
Net Profit After Tax	357	146	(182)	(23)	105
Profit Ratios					
Operating profit margin	39.45	22.18	(6.57)	36.68	40.77
Net profit margin	25.51	11.26	(52.95)	(5.00)	18.56
Leverage Ratio (D/E)	1.15	0.98	1.06	1.03	0.83
Debt to Capital Ratio	0.53	0.50	0.51	0.51	0.45
Return on Net Assets	0.38	0.11	(0.12)	(0.02)	0.06

Majorly AEGCL has been profitable; barring FY 20 and FY 21, wherein revenue had significantly decreased compared to FY 19.

Higher expenses in FY 18 and FY 19 attributable to significantly high Transmission charges paid to PGCIL as part of Other Expenses; while significantly higher revenue from Wheeling charges were earned in the same period.

9.1.4 Assam Power Distribution Company Limited (APDCL)

In pursuance of Notification Memo No. PEL151/2003/Pt./165 dated December 10, 2004 of the Government of Assam, three Distribution Companies were formed as a successor to the ASEB. Vide subsequent Notification No. PEL.41/2006/199 dated May 13, 2009, all these three Distribution Companies were merged into one and in accordance with the Assam State Reform (Transfer and merger of Distribution Functions and undertakings) Scheme, 2009 and Certificate of Incorporation dated October 23, 2009, the present Distribution Company, viz., Assam Power Distribution Company Limited (APDCL) got formed.

The main purpose of forming the Company was to take over, manage and operate the electricity distribution system, assets, liabilities, undertaking of the erstwhile Assam State Electricity Board (ASEB) pursuant to a notified transfer scheme in terms of Part XIII of the Electricity Act, 2003. Apart from Distribution, APDCL is also involved in power trading.

As would be detailed in subsequent sections, a substantial quantum of its power purchase is dependent on high-cost coal and gas-based generation. Further, APDCL also relies on short-term power purchase solutions which can also get very costly thereby potentially affecting the overall Average power purchase cost (APPC) of the DISCOM. Further, RPO targets have also increased significantly as set by the Ministry of Power. Thus, RE integration gives APDCL an opportunity to reduce its dependency on high cost power solutions and adopt sustainable energy options that would potentially help reduce their overall power purchase costs as well as help achieve RPO targets as well. It is however notable to highlight that APDCL also has a sizeable target pipeline for RE projects in the coming decade which if successfully implemented can significantly change the landscape of RE integration in state.

Either through floating tenders and purchasing RE power vide projects set up by Independent Power Producers (IPPs) or by executing the power projects independently or through JV, APDCL can play a key role in facilitating large scale RE integration in the state.

Financial Snapshot of APDCL[33]:

Table 11: Financial Snapshot - APDCL

Amount figures in INR Crores

Item	FY18	FY19	FY20	FY21	FY22	FY23*
Commercial Indicators						
Total Energy Sales (GWh)	6,814	6,968	7,257	7,458	8,520	9,043
Average Cost of Supply (₹/kWh)	7.85	8.72	8.97	9.36	8.73	10.87
Average Billing Rate (₹/kWh)	6.97	7.50	8.09	7.64	7.55	8.84
Financial Indicators (₹ Crores)						
Total Revenue	5,899	6,372	6,926	7,158	7,943	9,597
Total Expenses	5,740	6,374	7,039	7,451	7,835	10,229
Expenses (excluding Depreciation and Finance Cost)	5,426	6,009	6,650	6,956	7,278	9,607
Operating profit	473	363	277	202	666	(10)
Interest expenses	267	291	169	203	159	144
Gross Fixed Asset	3,824	4,436	5,779	7,931	8,437	NA
Net Fixed Asset	2,651	3,137	4,260	6,121	6,230	NA
Current Asset	6,431	5,920	5,165	4,238	4,459	6,008
Current Liability	6,339	6,502	5,743	5,136	4,827	7,696
Non-Current Liability	2,565	2,645	2,734	2,703	1,667	2,126
Total Equity	3,773	5,521	2,978	2,683	3,950	4,269
Depreciation and amortization	46	75	221	291	398	478

[33] APDCL Annual Accounts

Item	FY18	FY19	FY20	FY21	FY22	FY23*
Exceptional Items	5	23	1,067	-	228	-
Gross profit	164	21	954	(292)	336	(632)
Tax	-	-	-	-	-	-
Net Profit After Tax	164	21	954	(292)	336	(632)
Profit Ratios						
Operating profit margin	8.01	5.70	4.00	2.82	8.38	(0.10)
Net profit margin	2.79	0.33	13.77	(4.09)	4.23	(6.59)
Leverage Ratio (D/E)	2.36	1.66	2.85	2.92	1.64	2.30
Debt to Capital Ratio	0.70	0.62	0.74	0.75	0.62	0.70
Return on Net Assets	0.06	0.01	0.26	(0.06)	0.06	NA

*Unaudited

Revenue steadily increased till FY 23 at a CAGR of ~8.5%, but expenses grew at a faster rate of ~10.1% (CAGR). Owing to influx of Exceptional items of value Rs ~228 Cr (reversal of interest on GoA loans), PAT was positive in FY 22. In FY 23, significantly higher power purchase cost have contributed to Loss, this gap may get adjusted based on approval of the Commission

9.1.5 Assam State Load Despatch Centre (SLDC)

The State Load Despatch Centre (SLDC) is the apex body constituted vide Section 31 of the EA 2003 and complies with the directions stipulated in Section 33 to ensure integrated operation of the power system in the State of Assam. SLDC, Assam was established in the year 1983 and is located at Kahilipara, Guwahati. SLDC, Assam is presently being operated by AEGCL.

9.2 Annexure – Detailed Features of Assam RE Policy 2022

Figure 39: Detailed Features of Assam RE Policy 2022[34]

S. No	Provision Aspect	Details
1.	**Target Capacity (Clause 9)**	Target of **1200 MW** RE capacity addition under RE policy period (i.e. by 31st Mar 2027). Break up of target is as follows: • **Grid Connected Solar Projects – 620 MW** o *Solar Park – 100 MW* o *Solar plants with sale of power to APDCL – 300 MW* o *Solar plants with sale of power other than APDCL – 50 MW* o *Solar Power under REC Mechanism – 50 MW* o *Solar Power plant in Agri sector – 100 MW* o *Captive Solar Power Plant – 20 MW* • **Grid Connected Rooftop Solar Projects – 300 MW** o *Industry with Storage – 100 MW* o *Residential – 100 MW* o *State Govt. Installations* • **Off-grid Solar Applications – 80 MW** o *Solar Pump – 25 MW* o *Mini/Microgrid solar projects, solar lighting, off-grid solar plant – 5 MW* o *State Govt. Installations – 50 MW* • **Other Renewable Energy Sources – 200 MW** o *Small Hydro – 25 MW* o *Pump Storage – 50 MW* o *Bio-Mass – 25 MW* o *Solid Waste – 100 MW*
2.	**Solar Parks (Clause 10.1)**	o APDCL shall act as Solar Power Park Developer (SPPD) for development of infrastructure and management of solar park in state o APDCL will formulate the Business modalities in respect of land allotment, sharing of costs o APDCL shall purchase 100% of the aggregate capacity of solar energy produced from all solar parks in state o State will promote setting up of solar parks on water bodies for sale of power to DISCOM on Tariff Based Competitive Bidding (TBCB) route

3.	**Grid-connected Rooftop Solar (RTS) Projects (Clause 10.2)**	o State Govt. may make budgetary provision for providing payment security in case Govt./Semi-govt./Govt. aided organizations/Govt. owned corporations/ Statutory bodies etc. implement RTS project through RESCO
		o APDCL to procure entire power generated from RTS grid connected project under Gross Metering
		o DISCOM to notify substation-wise surplus capacity which can be fed from RTS plants to the grid
		o Energy Accounting and Billing
		  o Under EXIM metering –
		    • The energy consumed during a month by the consumer from the grid shall be billed as per prevailing tariff schedule of tariff applicable for the category
		    • The amount due to the consumer on account of injection of solar energy to the grid shall be reckoned of the APPC rate. This amount shall be adjusted from the monthly electricity bill
		  o Under Gross Metering –

[34] Assam RE Policy 2022 (No. PEL 230-2021-138 dated 13th October 2022)

S. No	Provision Aspect	Details
		• DISCOM shall procure entire generated power at a tariff discovered through competitive bidding route • PPA with DISCOM for quantity of injected units for 25 years o Allowed capacity of RTS project at single location – Min - 1 kW; Max- 1000 kW o Individual consumer – capacity upto sanctioned load or contract demand as applicable o Cumulative capacity allowed at a particular Distribution Transformer – 80% of DT capacity • No inspection from the State Electrical Inspectorate is required for RTS plants up to 500 kVA capacity.
4.	**Solar Plants for sale of power to APDCL (Clause 10.3)**	• Selection of Solar Power Generators (SPGs) and the discovery of corresponding tariffs etc. shall be done by APDCL • List of water bodies shall be identified in consultation with concerned department, Govt. of Assam. The bidder shall be selected through open competitive bidding by a separate tender by APDCL. • APDCL would enter into long term PPA of 25 years with SPGs who are selected based on a competitive bidding process. • The state Government shall evolve a guaranteed payment security mechanism to the investors under this category of projects.
5.	**Solar Plants for sale of power to DISCOMs other than APDCL (Clause 10.4)**	o Intrastate and interstate open access to be promoted by Govt. o Quantum of sale of power shall be subject to transmission and evacuation capacity of network o **Deemed open Access** provision (for no response from SLDC to application within 21 days from date of such application) o **Provision for execution of Wheeling and Banking Agreement with DISCOM under this category under captive mode**
6.	**Development of solar plants under REC Mechanism (Clause 10.5)**	• SLDC Assam designated as nodal agency for accreditation of solar projects under REC mechanism • Minimum capacity of 250 KW or as specified by CERC/AERC from time to time • Modalities of implementation to be issued by SLDC Assam • Policy benefits like Wheeling, banking, electricity duty shall be applicable as per provisions of CERC/ AERC • Projects set out under Clause 10.2, 10.4, 10.6 of the RE Policy shall also be eligible under REC mechanism
7.	**Captive Solar Power Plant (Clause 10.6)**	• State Commission (AERC) to be certifying body with respect to grant of captive status • For solar captive project at Assam consumer premises, sale of excess power to DISCOM will be @75% of APPC rate • For solar captive project at Assam consumer premises, sale of excess power to Third Party under Open Access, there will be exemption on Cross Subsidy Surcharge and Additional surcharges for 5 years • Provision for execution of Wheeling and Banking Agreement with DISCOM
8.	**Grid connected solar plants in Agri sector (Clause 10.7)**	• The State shall promote development of grid connected solar power plant to solarize agriculture pumps to meet the irrigation needs within the premises of individual farmer under EXIM metering. The surplus energy may be sold to DISCOM of the APPC rate as notified by State Commission. • APDCL shall be the Nodal Agency for promotion and implementation of Grid connected Solar Water Pumping systems in the State.

S. No	Provision Aspect	Details
		• The relevant authority of the concerned department (viz. Agriculture, Irrigation, Horticulture etc.) shall quantify the number of Grid Connected Agriculture Pumps to be energized through solar power in the State and same shall be intimated to Nodal Agency at the beginning of every financial year. • The State shall promote development of grid connected solar power plant on Barren uncultivable land or Agricultural land provided that solar plants to be Installed in stilt fashion (i.e. raised structure for installation of Solar panels) and with adequate spacing between panel rows for ensuring that farming activity is not affected. (Allowed capacity 500 kW – 2 MW)
9.	**Promotion of setting up of Renewable Energy based Electric Vehicle Charging Stations (Clause 10.8)**	State will promote the use of renewable energy for charging of EVs in the following manner: • The Charging Infrastructure will be developed as per the guidelines and standards issued by Ministry of Power and Central Electricity Authority. • The EV charging stations may be established by the State/Central Public Sector Undertakings, private operators or under public private partnership models. • The charging station service providers may set up solar power plants within their premises for captive use, and may also draw solar power through open access from generation plants located within the State to avail the benefits as provided under this Policy.
10.	**Promotion of Other Renewable Energy Sources (Clause 10.9)**	• Promotion of technologies like Pumped Storage, Biomass, Waste to Energy etc by State
11.	**Transmission and Distribution Network (Clause 11.2)**	For augmentation of transmission / distribution systems to evacuation the power from receiving sub-station, STU/DISCOM shall develop / augment the necessary transmission distribution network within mutually agreed timeframe.

12.	**Incentives (Clause 16)**	• With respect to grid connected RTS projects, without storage, excess generation would be purchased by APDCL at 75% of latest available bid tariff discovered by APDCL and same for 100% w.r.t RTS grid connected projects with storage
		• Exemption on Transmission and Wheeling charges as follows:
		  o *Transmission and wheeling charges are exempted upto March 2026 for projects wrt solar parks, grid connected RTS, solar projects for sale of power to APDCL, captive projects set up in Assam and selling to APDCL*
		  o *50% exemption of Transmission and wheeling charge for 3 years w.r.t Captive projects set up in Assam selling power to Third Party consumer in Assam and third party open access projects*
		  o *100% exemption on RES project for EV Charging infrastructure*
		• Exemption w.r.t Cross Subsidy Surcharge and Electricity Duty is as follows:
		  o *100% Exemption for 5 years w.r.t captive solar project with sale of power to third party within state, captive solar projects with storage, Third party open access project (within state)*
		• Electricity Duty and cess:
		  o *Exemption for 5 years with respect to grid connected RTS project, captive solar project selling power to APDCL or third party within Assam, solar projects with storage for captive use/third party sale, RE project for EV charging*
		• Deemed Open Access and banking provisions as mentioned earlier

S. No	Provision Aspect	Details
		• No clearance from any district authority or state authority or department is required on setting up of solar plants under the policy on lands or property otherwise not barred by any Act, rule or Judicial or executive order. • Grid connected Solar PV Projects will be exempted from obtaining any NOC/ Consent for establishment and operation under pollution control laws from Assam Pollution Control Board.

9.3 Annexure – Detailed Features of Assam RE Tariff Regulations 2017

Figure 40: Detailed Features of Assam RE Tariff Regulations 2017[35]

S. No	Provision Aspect	Details
1.	**Useful Life of Project**	25 Years
2.	**Technology Eligibility**	Based on Technologies approved by Ministry of New and Renewable Energy, Govt. of India
3.	**Tariff Structure**	• Return on equity • Interest on loan capital • Depreciation • Interest on working capital • Operation and maintenance expenses
4.	**Tariff Design**	• Levelized basis considering tariff from COD • For the purpose of levelized tariff computation, the discount factor equivalent to Post Tax weighted average cost of capital shall be considered
5.	**Debt-Equity Ratio**	• 70:30 • If equity is more than 30%, excess shall be treated as normative loan • **If equity is less than 30%, actual equity shall be considered for tariff**
6.	**Loan Tenure**	13 Years
7.	**Interest Rate**	• **2% + SBI MCLR Rate** (one year tenor) prevalent during the last available six months shall be considered.
8.	**Depreciation**	• Depreciation rate: **5.28% for 13 years** (rest to be spread across remaining 12 years so that maximum depreciation is 90%)
9.	**Return on Equity**	• The normative Return on Equity shall be **14%**, to be grossed up by prevailing Minimum Alternate Tax (MAT) as on 1st April of previous year for the entire useful life of the project

10.	**Interest on Working Capital**	• Working Capital Requirement:
		  ○ O&M expense for 1 month
		  ○ Receivables of 2 months for sale of electricity based on normative CUF
		  ○ Maintenance spare @ 15% of O&M expenses
		• Interest rate: 3% + SBI MCLR Rate (one year tenor) prevalent during the last available six months shall be considered.

11.	**O&M Expenses**	O&M expense will be project specific
12.	**Auxiliary Consumption**	0.25% of gross generation
13.	**Taxes and Duties**	• Tariff determined under these regulations shall be exclusive of taxes and duties as may be levied by the appropriate Government • Provided that the **taxes and duties levied** by the appropriate Government **shall be allowed as pass through on actual incurred basis** subject to production of documentary evidence by generating Company

[35] Source: https://aerc.gov.in/regulations/1671256076.pdf

9.4 Annexure - RE Policy Comparison

9.4.1 RE Policy Comparison – Cost Reduction Measures/Incentives

Table 12: RE Policy Comparison of Different States in India

Policy Focus	Karnataka RE Policy 2022	Odisha RE Policy 2022	Jharkhand Solar Policy 2022	UP Solar Policy 2022	MP RE Policy 2022	West Bengal RE policy 2012	Meghalaya a Power Policy 2007	Assam RE Policy 2022
Waiver of Transmission Charge and Wheeling Charge	As determined by KERC from time to time.	For intrastate: 25% and 50% waiver on wheeling and CSS charges respectively for 15 years Exemption of 20 paisa/Unit on STU charge for 15 years	For intrastate: • 100 % waiver for the period of 10 years from COD date • Cross Subsidy and Additional Surcharge exemption for sale to third party within state	For intrastate: • 100 % waiver if power sold to UPPCL; 50% waiver for sale to 3rd party within state • For interstate: 100 % waiver wheeling & Transmission Charge, CSS on STU network	The incentives shall include • rebate/ exemption on duties and cess, grant towards waiver of wheeling charges • Carbon credits or any other similar incentives, which are available for such Projects, can be availed by the Developer, as per the guidelines issued by the concerned authorities from time to time.	No waiver on OA charges	• No waiver on OA charges. • Provision for wheeling in state for captive use or to third party (HT consumer)	• 100% waiver of intrastate transmission and wheeling charges for Grid-connected rooftop, captive and private solar projects selling power exclusively to APDCL • 50% waiver for 3 years for Third party open access projects (sale within state) • 100% exemption for RE power sourced towards EV charging Station for 10 years • 100% exemption of CSS and Additional

Policy Focus	Karnataka RE Policy 2022	Odisha RE Policy 2022	Jharkhand Solar Policy 2022	UP Solar Policy 2022	MP RE Policy 2022	West Bengal RE policy 2012	Meghalaya Power Policy 2007	Assam RE Policy 2022
								Surcharge for Third party open access/captive solar
Banking Facility	• Facility available. as determined by KERC from time to time • No banking for ISTS	Banking facility available within State on monthly basis	100% banking facility available	Up to 100% for 25 years	NA	No provisions	No provisions	Allowed as per provisions determined by AERC from time to time
Exemption on Electricity Duty	• No exemption • on explicitly mentioned • Applicability of duty as per the direction of GoK issued from time to time.	Exemption of 50 Paisa/Unit for energy sold within State for 15 years (20 years for projects commissioned before 30-Mar-26)	• Banking and electricity duty charge exempted for 10 years • Exemption applicable for projects commissioned as per schedule	Exempted on energy sold to distribution companies / third party / captive use for initial 10 years	100% Exemption for period of 10 years.	No waiver	Exempted for 5 years with respect to captive use or third party open access	• 100% exemption of electricity duty and cess for 5 years from COD wrt following: • Grid connected rooftop solar • Captive sale to APDCL and Third party sale within state • Solar projects with storage for third party sale/ captive use • Solar Power sourced (including captive) for EV charging infra

Policy Focus	Karnataka RE Policy 2022	Odisha RE Policy 2022	Jharkhand Solar Policy 2022	UP Solar Policy 2022	MP RE Policy 2022	West Bengal RE policy 2012	Meghalaya a Power Policy 2007	Assam RE Policy 2022
Land related exemptions	NA	• 100% Exemption on Stamp Duty • Land conversion and registration charges exempted for RE projects	• 100% Exemption on Stamp Duty • 100% waiver on land use conversion charges/fees • Entry Tax not applicable • Exemption from land use approval, external development charges, scrutiny fee, infrastructure development charges and court fee related to leasing	• 100% Exemption on Stamp Duty • Deemed exemption from Land ceiling and change in land use.	50% reimbursement on stamp duty on purchase of private land for the project shall be available for developers.	No exemptions	No exemption s	No exemptions
Incentive/ Capital Subsidy	No incentives mentioned	Incentive applicable as per guideline of Central/State	For Rooftop Solar • 60% (upto 3kW if annual income less than INR 3 lakh) • 80% (3kW to 10 kW if annual income less than INR 3 lakh)	• Subsidy of INR 15000 per kW to the max limit of subsidy of INR 30,000 per consumer for Rooftop • INR 2.50 Crore per MW capital subsidy for utility scale projects + min. 4 hrs BESS (5 MW or above) and standalone BESS(energiz ed by solar energy only)	capital subsidy shall be provided to hawkers and street vendors on purchase of solar lanterns from agency empaneled with MPNRED.	Provision for subsidy related to off-grid solar and bio-mass projects.	No incentives mentioned	State subsidy of Rs 1000/kW upto max. of Rs 3000 per customer for grid connected rooftop solar projects for first 30 MW applications received in APDCL portal (over and above any CFA)

Policy Focus	Karnataka RE Policy 2022	Odisha RE Policy 2022	Jharkhand Solar Policy 2022	UP Solar Policy 2022	MP RE Policy 2022	West Bengal RE policy 2012	Meghalaya a Power Policy 2007	Assam RE Policy 2022
				for sale of power to Distribution licensee/ UPPCL				
Incentives for Solar Pumps	Support in implementation of MNRE schemes and programs	As per PM KUSUM Scheme	66-67% for Solar Pumps	State subsidy of 60-70 % in addition to 30 % subsidy available from GoI for various categories of farmers(SC/ST, Vantangia and Mushar category, others etc,)	NA	NA	NA	Excess generated power to be purchased by APDCL at APPC
Feeder Segregation	Support in implementation of MNRE schemes and programs	Promotion by state	30% of Project Cost	VGF of Rs 50 Lakh /MW available in addition to Rs 1.05 Crore /MW subsidy available from GOI	NA	NA	NA	NA
Incentive for Solar Manufacturing	As per State Industrial Policy	As per Odisha IPR (Industrial Policy Resolution)	• Waiver on Electricity Duty for 5 years • 100% exemption on stamp duty • SGST Reimbursement to Solar Manufacturer • 100% reimbursement of Custom Duty for 5 years	Incentive available under State MSME Policy and Uttar Pradesh Industrial Investment and Employment Promotion Policy	1.Incentives shall include concessional lease rent for government land, subsidies, rebate/exemption on duties etc. 2. RE equipment manufacturing units commissioned on or before 31st March 2027 shall be eligible to avail following benefits: a. Investment Size < 50 Cr.-	NA	NA	NA

Policy Focus	Karnataka RE Policy 2022	Odisha RE Policy 2022	Jharkhand Solar Policy 2022	UP Solar Policy 2022	MP RE Policy 2022	West Bengal RE policy 2012	Meghalaya Power Policy 2007	Assam RE Policy 2022
					Eligible to avail general incentives as per respective policy of Industry/MSME Department based on investment size. b. Investment Size > 50 Cr.- Eligible to avail special incentives earmarked for RE Equipment Manufacturing Sector under Industrial Promotion Policy			
Other Incentives	All RE projects eligible for additional incentives as per State Industrial Policy (and also proposed for availing benefits of manufacturing industry)	• No free power for sale/consumption of entire power from hydro project in state • Provision for formulation of VGF scheme for Pumped Storage hydro (except for captive or 3rd party sale)	Solar Parks – Cost of transmission till nearest substation to be borne by govt. (till 10 km for project on private land)	State will bear the cost of construction of Transmission Line upto 20 Km from project site for grid connectivity of Standalone solar power systems	Incentives on case-to-case basis shall be made available to biogas plant developed by Community/Resident Welfare Society/Gram Panchayats etc.	Cost of connectivity to be borne by Transco/Discom (as the case may be) for small-hydro and island-based wind projects, if the developer has to lay a line for a distance greater than 5 km, subject to Regulations of the Electricity Regulatory Commission	NA	Purchase of excess power by APDCL as per following: • @75% latest available bid tariff for grid connected rooftop • @100% latest available bid tariff for grid connected solar projects with storage

9.4.2 RE Policy Comparison – EODB Measures

Policy Focus	Karnataka RE Policy 2022	Odisha RE Policy 2022	Jharkhand Solar Policy 2022	UP Solar Policy 2022	MP RE Policy 2022	West Bengal RE policy 2012	Meghalaya Power Policy 2007	Assam RE Policy 2022
Approval Process	Concerned department shall convey the approval/ clearance/ observation or comments if any within a stipulated time period of 45 days from the date of application	• Single window facility for approval of all RE projects in the State	Single window approval for all type of Solar Projects	Single window approval for all type of Solar Projects	Establish a "Single Window System" for technical support, and project clearance through between the concerned Government department.	West Bengal Green Energy Development Corporation Limited (WBGEDCL) is the nodal agency acting as Single Window for obtaining assistance from all line Departments for clearances	• Simple composite approval form to MNREDA and DISCOM • Clearances to be granted within 2 months • Agreements will be signed within 1 month post according of approvals	No clearance from any district authority or state authority or department is required on setting up of solar plants under the policy on lands or property otherwise not barred by any Act, rule or Judicial or executive order.
Floating / Canal Top Solar	• GoK to promote floating solar on existing dams/reservoirs/water bodies • Guidelines for implementation to be notified	• Dept. of Water Resources would allow development of plants in its water bodies at nominal annual lease rent/ payment	• WRD along with the state GENCO's will provide the access to the water bodies or the dams to the developer • JREDA to be single window for clearance and would issue guidelines	• Floating solar will be promoted • To be implemented through competitive bidding	NA	NA	NA	• APDCL to procure at least ~50 MW power from floating solar projects • List of water bodies shall be identified
Land	• Deemed NA conversion status If	• IDCO's land under "Land	• Nodal agency,	• Nodal Agency to create land bank	NA	• Provision for lease of	For Hydro: The State	Exemption from

Policy Focus	Karnataka RE Policy 2022	Odisha RE Policy 2022	Jharkhand Solar Policy 2022	UP Solar Policy 2022	MP RE Policy 2022	West Bengal RE policy 2012	Meghalaya Power Policy 2007	Assam RE Policy 2022
	SLAC/Energy Department of Karnataka gives the approval • No other Project approval required (excluding applicable statutory approvals) from any other Government departments	Bank" Scheme and other Govt Land to be provided for RE at IPR rates on priority basis • Nodal Agency shall work in coordination with the Dist. Collector /IDCO to identify land parcels for RE • Deemed NA conversion for RE projects	JREDA, will identify and aggregate State lands, private lands, and canal tops • JREDA to float EOI for private land • Land Bank to be created	specifically in Bundelkhand region • Procedural delays to be minimized		govt. vested land for 30 years subject to availability • No other incentives mentioned	Government shall expedite suitable approval under the Meghalaya Land Transfer Act so as to facilitate investment in the Power Sector.	requirement of Consent to Establish and Consent to Operate (CTE/CTO) under pollution control laws from Assam Pollution Control Board for development of grid-connected solar PV projects
Solar/RE Park	• GoK will promote RE parks under PPP model • Equity investment support (up to 50% or more) as and when decided upon by Govt.	• Solar Park by State entities: Any SPSU/CPSU either on its own or through a JV can execute solar park project • State will also promote Solar Park by Private SPPD	JREDA may establish a Special Purpose Vehicle (SPV) to develop the infrastructure in the solar park and to manage the solar park	Solar park on Govt Land: Land will be provided on lease or Right to Use basis for 30 years at Rs 1/- per acre per year.	NA	NA	NA	APDCL shall purchase 100% of aggregate capacity of solar produced from all solar parks in state
Other drivers	• Peer to Peer (P2P) trading of RTS energy to be promoted • The transmission evacuation approval by KPTCL is to be done by 60 days from applying	• Captive RE projects shall be provided approvals in fast-track mode. • Enabling provisions for	• Deemed open access in the absence of any intimation from	• State Level Empowered Committee (SLEC), Department level Empowered committee(DLEC) and District Level	NA	In case of RE project construction in very remote areas, some infrastructural support	Infrastructural facilities such as approach roads, water supply, crane, power during construction period etc. will	Provision for setting up of 1000 MW solar plants on free unused government land as per

Policy Focus	Karnataka RE Policy 2022	Odisha RE Policy 2022	Jharkhand Solar Policy 2022	UP Solar Policy 2022	MP RE Policy 2022	West Bengal RE policy 2012	Meghalaya Power Policy 2007	Assam RE Policy 2022
	• Green energy corridor will be formed in a PPP scheme • GoK will promote repowering of existing wind turbines which have completed at least 10 years in operation	"Just Transition" through capacity building • Projects in the Solar Parks shall be free to sell the entire saleable power within or outside the State or for captive use	Jharkhand DISCOMs • Unutilized banked energy shall be considered as deemed purchase by DISCOMs at 50% of APPC (capping of 10% of total banked energy)	Committee (DLC) for fast tracking approval mechanisms and policy implementation • Formation of Uttar Pradesh Solar Energy Development fund (UPSEDF) for enhancing institutional framework		including approach roads to the project site may be provided at Government cost	be provided on the lines of industrial estates.	"Mukhya Mantri Souro Shakti Prokolpo"

9.5 Annexure - Details of BOO based solar projects under SECI bidding

Table 13: List of Commissioned SECI Projects under BOO Model[36]

Sr. No.	Name of Project Developer	Project Capacity (MW)	Project Location (State)	Actual Commissioning Date	Total capacity commissioned as on date (MW)
Solar Tenders under SBG					
2000 MW ISTS Connected Solar PV Projects (ISTS-I)					
1	Mahindra Renewables Private Limited	**250**	Vill-Bhiv Ji ka Gaon, Tehsil-Bap, Dist.-Jodhpur	14.05.2021 (100 MW) 07.06.2021 (60 MW) 23.07.2021 (50 MW) 17.08.2021 (40 MW)	250
2	Kilaj Solar (Maharashtra) Pvt. Ltd	**50**	Vill:, Nure Ki Bhurj/Sihra, Dist: Jodhpur	17.Apr.2020	50
5	Azure Power Forty-Three Private Limited	**300**	Village Jagdev wala and Daudsar, Tehsil Bikaner, District Bikaner	15.12.2020 (150 MW) 30.01.2021 (100 MW) 08.02.2021 (50 MW)	300
6	Azure Power Forty-Three Private Limited	**300**	Village Jagdev wala and Daudsar, Tehsil Bikaner, District Bikaner	25.06.2021 (50 MW) 08.08.2021 (50 MW) 22.09.2021 (100 MW) 31.12.2021 (100 MW)	300
7	Clean Solar Power (Jodhpur) Pvt. Ltd	**250**	Vill-Noore Bhurj, Tehsil-Bap-Dist.-Jodhpur	14.03.2022 (115 MW) 21.04.2022 (135 MW)	250
750 MW Grid Connected in Rajasthan Tranche-I (non-solar park)					
2	Fortum Solar Plus Pvt. Ltd.	**250**	Vill: Bainsada , Tehsil: Pokhran, Dist: Jaisalmer, State Rajasthan	19.03.2021 (150 MW) 01.05.2021 (100 MW)	250
3	ReNew Solar Energy (Jharkhand Five) Pvt. Ltd	**110**	Vill: Bhainsara, Tehsil: Fatehgarh, Dist: Jaisalmer, State Rajasthan	04.02.2021 (50 MW) 11.03.2021 (60 MW)	110
4	Sitara Solar Energy Pvt. Ltd.	**100**	Vill: Kanasar, Tehsil: Bap, Dist: Jodhpur, State Rajasthan	20.May.2021	100
5	Palimarwar Solar House Pvt. Ltd.	**20**	Vill: Hariyasar, Tehsil: Pokhran, Dist: Jaisalmer, State Rajasthan	19.04.2021 (20 MW)	20
6	Palimarwar Solar House Pvt. Ltd.	**20**	Vill: Hariyasar, Tehsil: Pokhran, Dist: Jaisalmer, State Rajasthan	19.04.2021 (20 MW)	20
1200 MW ISTS Connected Projects (ISTS-III)					
1	SBSR Power Cleantech Eleven Pvt. Ltd.	**300**	Vill: Lalsar & Hapasar, Tehsil: Bikaner, Dist: Bikaner, State Rajasthan	15.08.2021 (50 MW) 04.04.2022 (50 MW) 11.04.2022 (50 MW)	150
2	ReNew Sun Waves Pvt. Ltd.	**300**	Vill: Mandhopura, Tehsil: Fatehgarh, Dist: Jaisalmer, State Rajasthan	13.08.2021 (150 MW) 17.08.2021 (50 MW) 04.09.2021 (50 MW) 04.10.2021 (50 MW)	300

[36] Source: SECI

Sr. No.	Name of Project Developer	Project Capacity (MW)	Project Location (State)	Actual Commissioning Date	Total capacity commissioned as on date (MW)
Solar Tenders under SBG					
3	Eden Renewable Cite Pvt. Ltd.	**300**	Vill:Lakhasar, Tehsil: Pokhran, Dist: Jaisalmer, State Rajasthan	13.08.2021 (300 MW)	300
4	Azure Power Forty One Pvt. Ltd.	**300**	Vill: Noore Ki Bhurj, Khakhuri, Dedasari, Kushla Ram ki Basti, Ismail ki Dhani, Tehsil: Bap, Dist: Jodhpur, Rajasthan	12.10.2021 (50 MW) 30.10.2021 (50 MW) 30.11.2021 (50MW) 24.12.2021 (50 MW) 31.01.2022 (50 MW) 07.03.2022 (50 MW)	300
1200 MW ISTS Connected Projects (ISTS-IV)					
1	Ayana Renewable Power One Pvt. Ltd.	**300**	Vill: Khichiya, Bikaner Dist., Rajasthan	25.11.2021 (100 MW) 07.12.2021 (50 MW) 22.12.2021 (150 MW)	300
2	Renew Solar Energy (Jharkhand Three) Pvt. Ltd.	**300**	Vil: Mandhopura, Tehsil: Fatehgarh, Dist Jaisalmer, Rajasthan	02.09.2021 (150 MW) 27.09.2021 (50 MW) 04.10.2021 (50 MW) 09.12.2021 (50 MW)	300
3	Azure Power Maple Pvt. Ltd.	**300**	Vill: Sonanda, Shekhar, Bandhari & Kesarapura Teshsil: Bap, Dist. Jodhpur, Rajasthan	14.02.2022 (53 MW) 30.03.2022 (204 MW)	300
4	Mega SuryaUrja Pvt. Ltd.	**250**	Jodhpur Dist., Rajasthan	20.05.2022 (175 MW)	250
750 MW Grid Connected in Rajasthan Tranche-II (non-solar park)					
1	NTPC Limited	**160**	Vill: Jetsar, Dist. Sri Ganga Nagar, Rajasthan	22.10.2021 (80 MW) 25.03.2022 (80 MW)	160
2	Mahindra Susten Pvt. Ltd.	**200**	Bikaner Dist. Rajasthan	14.10.2021 (200 MW)	200
3	Clean Solar Power (BHAINSADA) Pvt. Ltd.	**250**	Vill: Bhainsada, Amarsar/Balwantgarh, Tehsil: Bhaniyana, Dist: Jaisalmer, Rajasthan	05.04.2022 (150 MW) 05.05.2022 (100 MW)	250
1200 MW ISTS Connected Projects (ISTS-V)					
1	GRT Jewellers (India) Pvt. Ltd.	**150**	Taluka: Ettayapuram, Kayathar, District: Thoothukkudi, Tamil Nadu	30.03.2022 (50 MW) 13.04.2022 (50 MW)	150
1200 MW ISTS Connected Projects (ISTS-VI)					
1	Avaada Sustainable RJProject Private Limited	**300**	Village: Noorsar, Tehsil: Bikaner in Bikaner, Rajasthan	14.01.2022 (100 MW) 07.02.2022 (50 MW) 12.04.2022 (50 MW) 11.05.2022 (100 MW)	300
2	Masaya Solar Energy Private Limited	**300**	Khandwa, Madhya Pradesh		150
3	ReNew Solar Urja Private Limited	**300**	Village Mandhopura, Tehsil Fatehgarh, District Jaisalmer, State Rajasthan	01.12.2021 (150 MW) 07.12.2021 (50 MW) 13.12.2021 (50 MW) 16.12.2021 (50 MW)	300
2000 MW ISTS Connected Projects (Tranche-IX)					

Sr. No.	Name of Project Developer	Project Capacity (MW)	Project Location (State)	Actual Commissioning Date	Total capacity commissioned as on date (MW)
Solar Tenders under SBG					
1	Thar Surya 1 Pvt. Ltd.	**300**	Vill: Dholera 1, Jamsar, Dist: Bikaner, Rajasthan	23.03.2022 (54 MW) 23.06.2022 (54 MW) 09.08.2022 (54 MW) 11.10.2022 (74 MW) 24.11.2022 (64 MW)	300

Note: ~5.6 GW projects commissioned as on August 2023 (excl. CPSU projects under SECI)

9.6 Annexure – Details of Solar Park (BOOT) based solar projects across states

Table 14: State Wise Details of Solar Parks[37]

#	State	Name of Solar Park	Approved Capacity (MW)	Status
1	Andhra Pradesh	Ananthapuramu-I Solar Park	1400	Completed
2		Kurnool Solar Park	1000	Completed
3		Kadapa Solar Park	1000	
4		Anathapuramu-II Solar Park	500	Completed
5		Solar-Wind Hybrid Solar Park	200	
6	Chhattisgarh	Rajnandgaon Solar Park	100	
7	Gujarat	Radhanesda Solar Park	700	
8		Dholera Ph-I Solar Park	1000	
9		NTPC RE Park	4750	
10		GSECL RE Park	3325	
11		GIPCL RE Park Ph-I	600	
12		GIPCL RE Park-Ph-II	1200	
13		GIPCL RE Park-Ph-III	450	
14	Himachal Pradesh	Kaza Solar Park	880	
15		Kinnaur Solar Park	400	
16	Jharkhand	Floating Solar Park	100	
17		Deogarh Solar Park	20	
18		Palamu Solar Park	20	
19		Garwha Solar Park	20	
20		Simdega Solar Park	20	
21	Jharkhand & West Bengal	DVC Floating Solar Park Ph-I	755	
22		DVC Floating Solar Park Ph-II	234	
23	Karnataka	Pavagada Solar Park	2000	Completed
24		Kalburgi Solar Park	500	
25	Kerala	Kasargod Solar Park	105	
26		Floating Solar Park	50	
27	Madhya Pradesh	Rewa Solar Park	750	Completed
28		Mandsaur Solar Park	250	Completed
29		Neemuch Solar Park	500	
30		Agar Solar Park	550	
31		Shajapur Solar Park	450	
32		Chhatarpur Solar Park	950	
33		Omkareshwar Floating Solar Park	600	
34		Morena Solar Park	1400	
35		Barethi Solar Park	630	

[37] Source: Lok Sabha Unstarred Question No. 440 for 08.12.2022

#	State	Name of Solar Park	Approved Capacity (MW)	Status
36	Maharashtra	Sai Guru Solar Park	500	
37		Dondaicha Solar Park	250	
38	Mizoram	Vankal Solar Park	20	
39	Odisha	Solar park by NHPC	40	
40		Floating Solar Park Ph-I	100	
41		Floating Solar Park Ph-II	200	
42	Rajasthan	Bhadla Phase-II Solar Park	680	Completed
43		Bhadla III Solar Park	1000	Completed
44		Bhadla IV Solar Park	500	Completed
45		Phalodi-Pokaran Solar Park	750	
46		Fatehgarh Phase 1B Solar Park	421	
47		Nokh Solar Park	925	
48		Pugal Solar Park Phase-I	1000	
49		Pugal Solar Park Phase-II	450	
50		RVUN Solar Park	1310	
51	Uttar Pradesh	Solar Park in UP	365	
52		Jalaun Solar Park	1200	
53		Kalpi Solar Park	65	
54		Mirzapur Solar Park	100	
55		Lalitpur Solar Park	600	
56		Jhansi Solar Park	600	
57		Chitrakoot Solar Park	800	
			39285	

~39.28 GW approval for solar parks have been accorded out of which, ~8 GW solar parks have been completed.

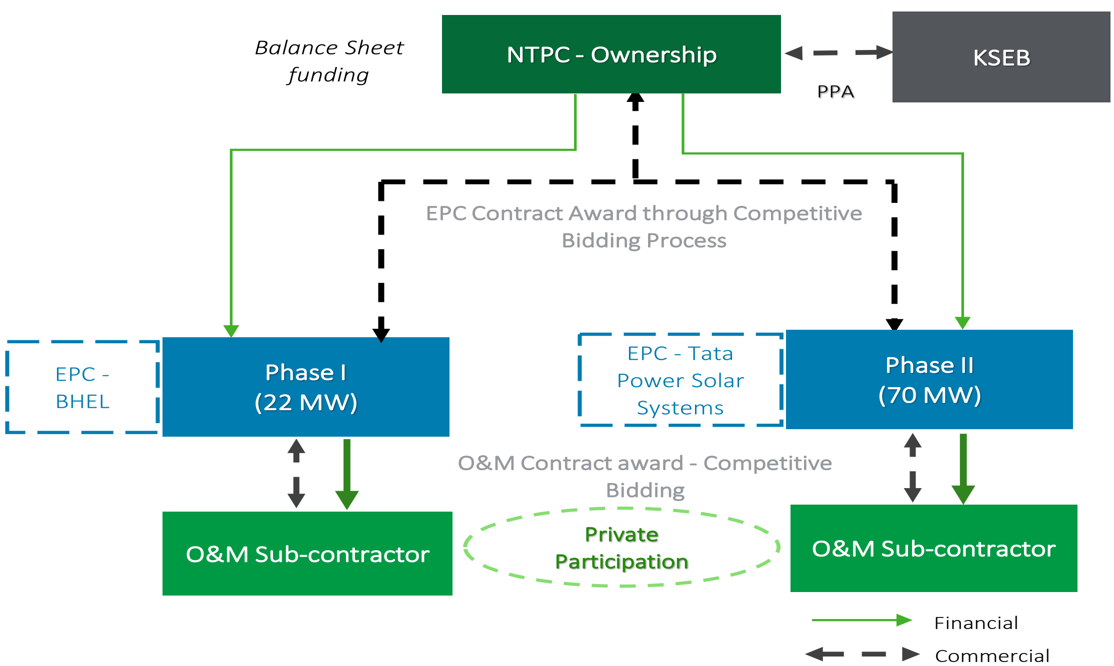
9.7 Annexure – Case Study (Public sector owned utility scale solar models)

9.7.1 Case Study: NTPC 92 MW Kayalkulam Floating Solar Plant (FSP), Kerala[38]

Project Details:

- MOU signed between the NTPC and the Kerala State Electricity Board (KSEB) in May 2018.
- NTPC initiated the bidding process for selection of EPC Contracts in two phases
- EPC cum O&M contracts awarded to two different companies
- Plant fully operational from June 2022

Figure 41: NTPC FSP (92 MW) at Kayalkulam



Key Highlights:

Table 15: Key Highlights of NTPC Kayakulam FSP Project

Aspect	Details
Project Capacity	• Phase I: 22 MW (EPC Contractor: M/S BHEL) • Phase II: 70 MW (EPC Contractor: M/S Tata Power Solar Systems Limited)
Capital Cost	Cost Approved by CERC • Phase I (22 MW): ~Rs 115 Cr • Phase II (70 MW): ~Rs 404 Cr
Funding	• Balance Sheet Financing by NTPC • D/E ratio of 80:20 as considered by CERC
Commercial	• PPA with KSEB at Rs 2.94 per unit • Tariff determined u/s 62 of EA 2003

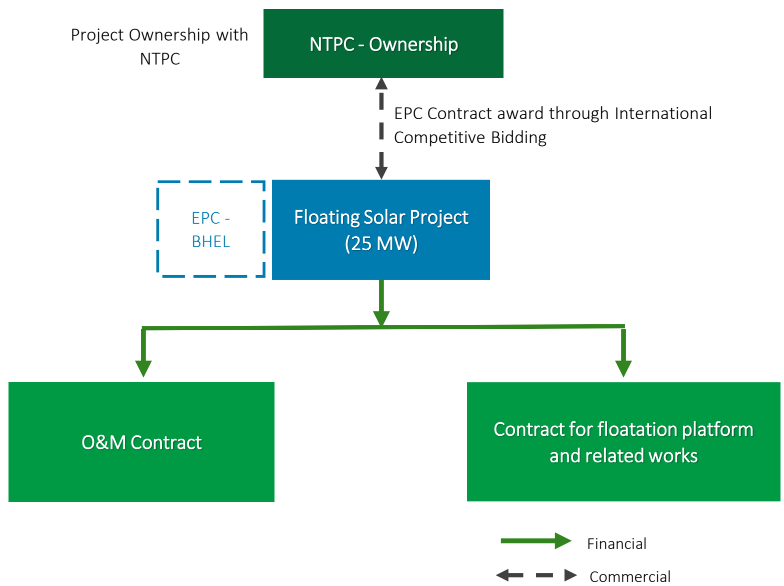
[38] Source: Deloitte Research

9.7.2 Case Study: O&M and EPC contract NTPC 25 MW Simhadri Floating Solar Plant, Andhra Pradesh[39]

Project Details:

- Project executed by NTPC under Flexibility in Generation and Scheduling of TPS to reduce emissions scheme of MoP(05-Apr-2018) to replace the thermal power supplied from Simhadri STPS to its beneficiaries
- Commissioning in two phases: 10 MW on 30-Jun-2021 and 15 MW on 21-Aug-2021
- Net gain from the supply of solar power in place of thermal power shared with the beneficiaries in the ratio of 50 (Beneficiary): 50 (NTPC)

Figure 42: NTPC Simhadri Floating solar project framework



Key Highlights:

Table 16: Key Highlights of NTPC Simahdri FSP Project

Aspect	Details
Project Capacity	• 25 MW (EPC Contractor: M/S BHEL)
Capital Cost	• Cost Approved by CERC • INR 100 Crore ($ 14.57 Mn)
Commercial	• No separate agreement and replacement of power to be considered within the existing PPA • Provision for sale of power (at ECR of Simhadri STPS) in open exchange • Landed price of RE power < Energy Charges of generating station whose power is replaced

[39] Source: Deloitte Research

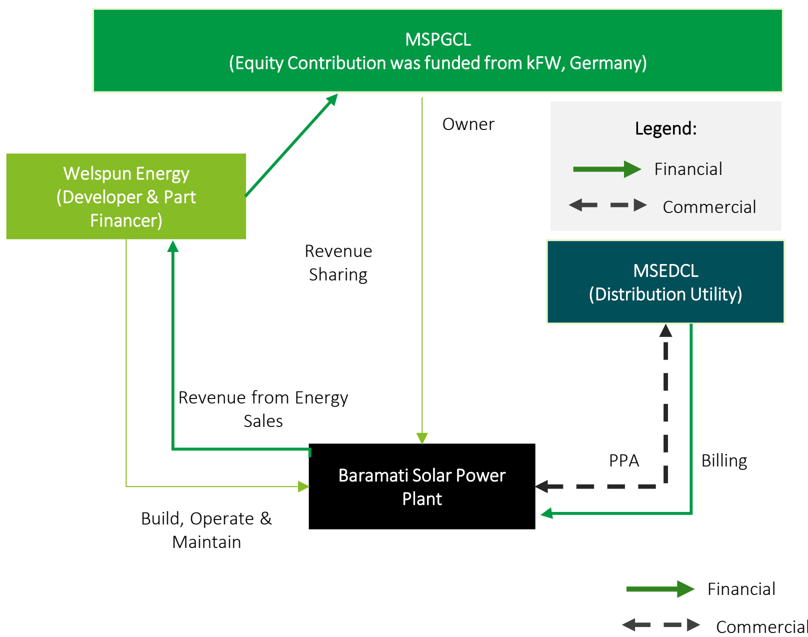
9.8 Annexure - Utility Owned Model – Private Participation - Mahagenco Case Study:

A Case Study of Utility Owned Model with Private Participation has been enumerated below:

Project Details:

- Maharashtra State Power Generation Company Limited (MSPGCL / Mahagenco) owns a 50 MW Grid Connected Solar Project in Baramati Maharashtra
- The project was a unique PPP model in India as it involved revenue share with the project owner for 25 years (linked to PPA)
- Welspun India developed the project on a Build, Operate and Maintain basis with MSPGCL

Figure 43: Project Overview



Key Highlights:

Table 17: Key Highlights of Mahagenco Project

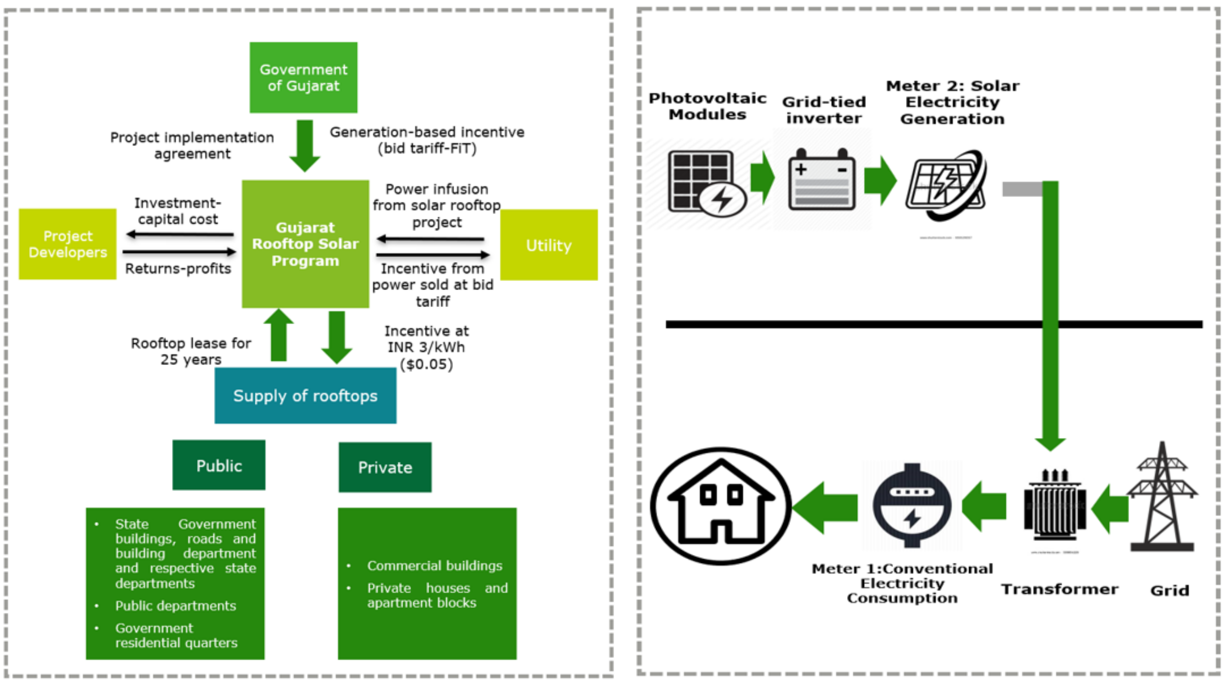
Aspect	Details
Project Funding	• Jointly funded by MSPGCL (Owner) & Welspun (Developer)
Mode of selection of Developer	• The successful bidder was selected based on the net present value of the highest revenue share offered to MSPGCL over the life of the project.
Power Purchase Agreement	• PPA with MSEDCL at Tariff Rate discovered via Bidding process
Revenue Sharing	• The bidder with the highest revenue sharing with MSPGCL was offered the contract. • Thus Welspun offered 38.75% revenue sharing and won the project • The revenue share was offered assuming CUF of 19%

9.9 Annexure - Distributed/Rooftop Solar Models

9.9.1 PPP based Gross Metered Model – Gujarat

In a gross metering model, entire generation of solar rooftop project is fed into the grid and sold to utility at the DISCOM offered rate. Rooftop owner does not consume the generated solar energy directly. A PPP based gross-metered Rooftop solar model adopted in Gujarat is presented below:

Figure 44: PPP based Gross-metered Solar rooftop model

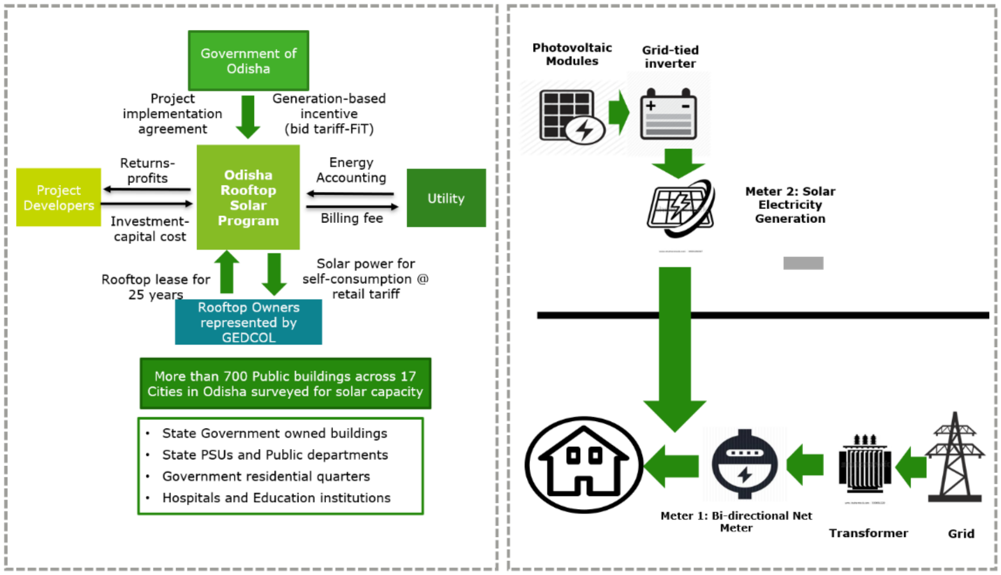


Private developer was selected through tender process. To mitigate the risk of rooftop availability, the government committed to lease public building rooftops to meet at least 80 percent of total capacity. A project development and management unit was designated to undertake a detailed technical, commercial, and regulatory assessment for the project.

9.9.2 PPP based Net Metered Model – Odisha

In a net metering model, a bi-directional meter is installed along with the Rooftop solar project. The rooftop owner can consume the power generated from the Rooftop plant. In a net metered arrangement, excess generated power from solar plant can be fed into the grid through the net meter and during non-solar hours, power would be sourced from the grid again through the same meter. The net energy exported or imported would be factored in while utility would bill the consumer. PPP also potentially finds application here as is illustrated through the PPP based net metering project adopted in Odisha as presented below.

Figure 45: PPP-based Net metered Rooftop Solar Model

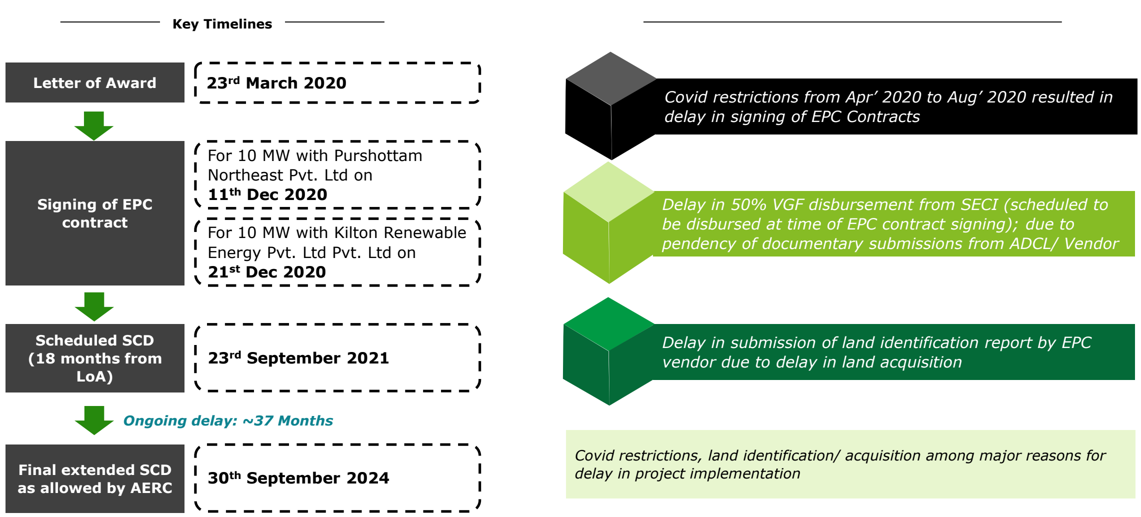


Government of Odisha partnered with the private sector to finance and set up solar panels on almost 200 buildings in the cities of Bhubaneswar and Cuttack with a minimum capacity of 4 MV. Transaction advisory was appointed by State-owned Green Energy Development Corporation of Odisha Limited, to assist in reviewing, structuring and implementing the project. A "Build Own Operate" (BOO) model was adopted here wherein, the winning bidder will set up solar panels across 200 Government and public sector buildings, with a minimum 4MW of installed capacity in Bhubaneswar and Cuttack. The lease agreement for Phase I of the project (200 KW) has been signed, and the design approved by the Independent Engineer.

9.10 Annexure – Solar Project Specific Issues Faced in Assam

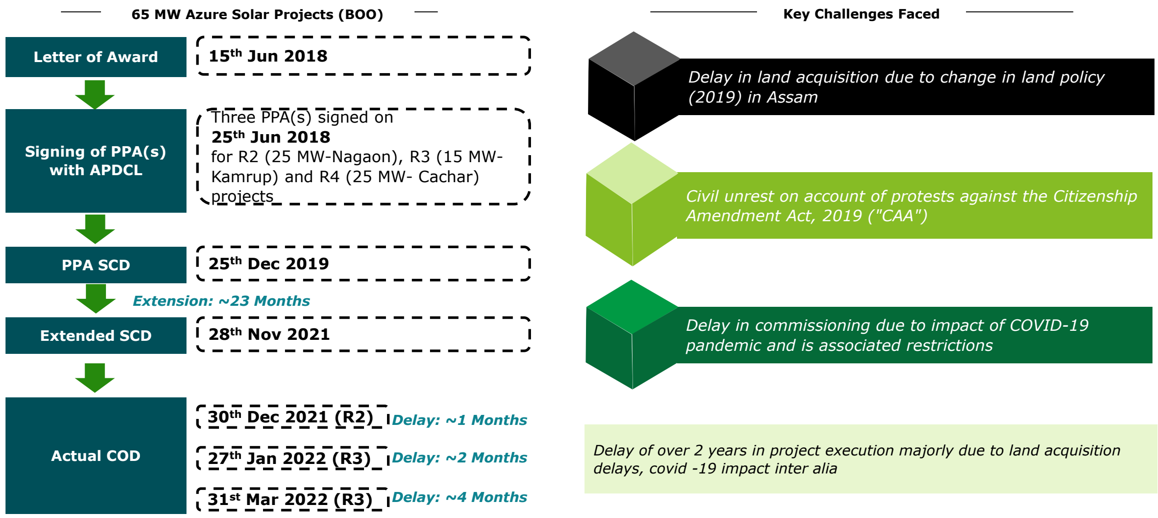
9.10.1 Issues faced in implementation of APDCL 2 X 10 MW projects under CPSU scheme

Figure 46: Issues faced in implementation of APDCL 2 X 10 MW projects under CPSU scheme



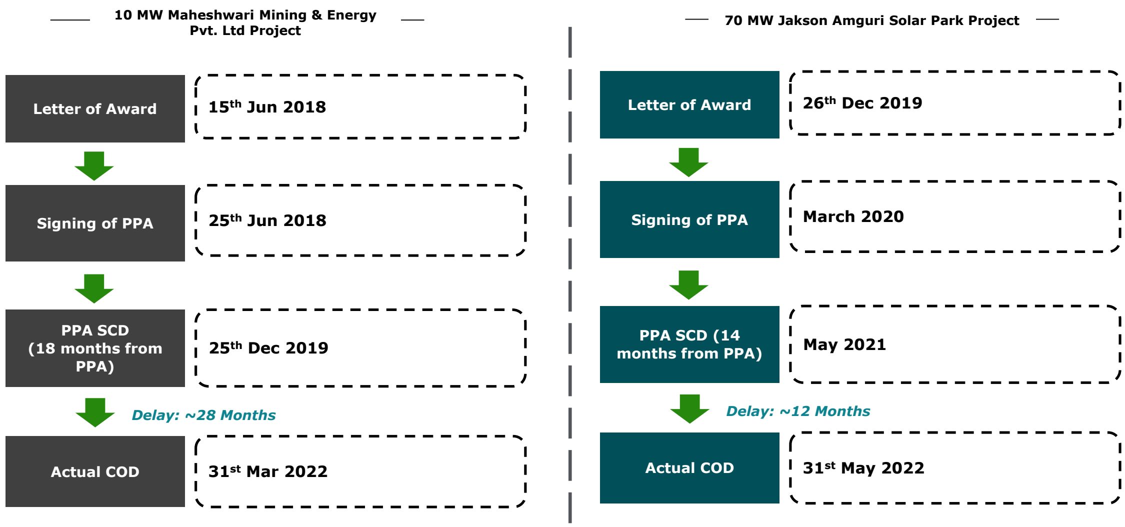
9.10.2 Issues faced in implementation of commissioned 65 MW Azure Power projects

Figure 47: Issues faced in implementation of commissioned 65 MW Azure Power projects



9.10.3 Key Timelines in project execution for some other solar projects in Assam

Figure 48: Key Timelines in project execution for some other solar projects in Assam



9.11 Annexure – List of Attendees in Stakeholder Workshop for Solar Roadmap in Assam held on 20th September 2023

Sl No.	Name	Designation	Department/ Organization
1.	Sh. Niraj Verma	ACS	Power Department, Govt. of Assam
2.	Sh. Rakesh Kumar	Managing Director	APDCL
3.	Dr. Jeevan B.	Secretary to Hon'ble CM, Govt. of Assam	
4.	Sh. Adil Khan	Secretary, Transport Department, Govt. of Assam	
5.	Sh. Lakshmanan S.	Secretary, Industries, Commerce & PE Department, Govt. of Assam	
6.	S. N. Kalita	Member, Technical	AERC
7.	Tushar Sud	Partner	DTTILLP
8.	Samrat Roy	Director	DTTILLP
9.	Siddharth Mani	Associate Director	DTTILLP
10.	Adit Khiroria	Manager	DTTILLP
11.	Rajeswar Banerjee	Sr. Consultant	DTTILLP
12.	D. C. Das	GM	NRE, APDCL
13.	Baishali Talukdar	Dy. Manager	NRE, APDCL
14.	B. Shiva	Manager	MMEPL
15.	Pranab Kumar Bordoloi	CGM (D&S)	APDCL
16.	Jiwan Acharya	Energy Specialist	ADB
17.	B. Das	Dy. Manager	APDCL
18.	Biswajit Biswas	VP Solar	Better Power Services Power Ltd.
19.	Amar Chetri	AGM	APGCL
20.	P. Nath	AGM	APDCL
21.	Jatin N.	DGM	APDCL
22.	S. Prasad	Jr. Engineer	MMEPL
23.	Paul White	ADB Consultant	ADB
24.	P. Roy Chowdhury	CGM	APDCL
25.	Dr. Jaideep Baruah	Director	AEDA
26.	B. Sinha Choudhary	CGM (NRE)	APDCL
27.	Sumit Kr. Singh	AGM	APDCL
28.	Bhaskar Jyoti Nag	AGM	APDCL
29.	Madhurjya Neog	DGM	APDCL
30.	Dhawal Jhamb	PPP Specialist	ADB
31.	Pritam Khanikar	AGM	APGCL
32.	S. Das	Jr Manager	APDCL
33.	Panchamrita Sharma	JD (T&RA)	AERC
34.	Arnab Ghosh	AGM (NRE)	APDCL
35.	Anindita Baruah	Jr Manager	APDCL
36.	Ranjan Goswami	AGM	AEGCL
37.	Kartikey Singh	Exe.	OTPC
38.	Puneet Khandelwal	Head, Energy	WRII
39.	B. S. Bedi	Procurement Consultant	ADB
40.	Akshay Rudra	BDM	L&T
41.	Rajesh Gupta	Head-BD	L&T
42.	Nipen Kr. Deka	Jt. Director (Engg.)	AERC
43.	Hemanga Deka	Scientific Officer	AEDA
44.	Kuldip Sarma	AGM	APDCL
45.	Debjani Biswas	DGM	APDCL
46.	Manash P. Kalita	AGM	APDCL
47.	Jitu Das	Consultant	APDCL
48.	Masfick Hazarika	Project Manager	WRII
49.	M. Dasgupta	CGM (F)	APDCL
50.	Jyotirmoy Banerjee	Energy Specialist	ADB
51.	Eva Borkakoty	DGM	APDCL
52.	R. Sarma	CGM	APDCL
53.	P. Kalita	AGM	APDCL
54.	S. Baruah	DGM	APDCL

Sl No.	Name	Designation	Department/ Organization
55.	A. Nasir	AGM	APDCL
56.	S. S. Bora	AGM	APDCL
57.	Putul Bhagowati	CGM	APDCL
58.	C. K. S. Bordoloi	GM	APDCL
59.	Abhijit Das	AGM (IT)	APDCL
60.	D. Hazarika	AGM (IT)	APDCL
61.	Partha P. B.	AGM	APDCL
62.	Partha P. Bharali	AGM	APDCL
63.	Debabrata Dutta	CGM(IT)	APDCL
64.	Lilambar Das	DGM (TRC)	APDCL
65.	Rajesh Payeng	DGM (CAR)	APDCL
66.	Guna Ram Saikia	DGM	APDCL
67.	Deepak Yadav	AGM (F&A)	APDCL
68.	Sathish	(FI)	Fichtner India
69.	Siva. R	DCE (FI)	Fichtner India
70.	Indrajit Tahbildar	AGM	APDCL
71.	Bhaskar Kalita	AGM	APDCL
72.	Bimala Brahma	CGM	APDCL